Engine Model(s): PT6A

72-00-00

ENGINE, TURBOPROP - FAULT ISOLATION

1. General

A. For details on recommended procedures for engine system fault isolation, reference should be made to the Fault Isolation Charts (Figures 101 through 105 and Table 101).

B. The charts comprise a series of diagnostic tests and rectification sequences which will assist an operator in discovering, isolating and correcting various faults that may arise in the basic engine or any of its related systems during service.

C. Engine faults can be either obvious or hidden. If hidden faults are not detected, serious and considerable damage may occur to the engine. Therefore, it is essential to have a thorough knowledge of correct turbine gas temperature and other important details of engine operation. Before attempting to locate a fault or difficulty, or to work on an engine which has been malfunctioning during flight, consult flight log and any other available data that could help in diagnosing the fault.

D. To correctly isolate a fault, check all previous information of engine faults, if any, and work that has been performed on the engine. Check each probable source of fault by use of relevant diagnostic test sequences until defect has been isolated. Systematic checking, essential for thorough fault isolation, will save time and extend engine life.

2. Consumable Materials

Not Applicable

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Engine Condition Trend Monitoring System

A. General

Gas turbine/turboprop engine maintenance practices frequently include in-flight engine performance monitoring as a means of detecting mechanical deterioration in engine gas paths. A simple system, requiring almost no arithmetic calculation, has been devised for the PT6A engine to aid in early planning of indicated rectifications, and thus reduce the potential costs of primary and secondary damage resulting from fully developed failures, or from the risks of in-flight shutdowns, and flight cancellations.

The turbine engine characteristic of repeatedly producing its output at, or very close to, charted gas generator parameter values provides the basis for the engine condition trend monitoring system. Thus, under known conditions of pressure altitude (PA) and indicated outside air temperature (IOAT), the gas generator parameter values of interturbine temperature

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(ITT), compressor speed (Ng), and fuel flow (Wf), for particular propeller shaft speeds (Np) and torques are predictable within reasonable limits.

New engines operate within a tolerance band, either above or below charted parameter values, and tend to deviate more from these values with time and deterioration of gas path components.

Sudden changes or gradually increased rate of change of the normal deviations from charted parameter values are very important indicators of gas path component conditions. As such, you can find the changes before some distress occurs.

The monitoring system must be introduced when engines are new or newly overhauled. This enables establishment of a performance base line before any deterioration takes place in the engine. For procedure refer to P&WC Publication SIL No. GEN-055.

6. Fault Isolation

A. Categories

To enable systematic fault isolation, engine faults are categorized into classes. Each class consists of common related engine problems with appropriate diagnostic tests and rectification sequences. The classes of engine fault and probable causes are detailed as follows:

1. Engine Starting (Ref. Subpara. B.)
2. Engine Operating (Ref. Subpara. C.)
3. Engine Performance (Ref. Subpara. D.)
4. Engine Condition Trend Monitoring Shift (Ref. Subpara. E.)
5. Engine Lubrication (Ref. Subpara. F.)
6. Engine Lubrication Oil Contamination (Ref. Subpara. G.)

B. Engine Starting

(1) To diagnose and rectify engine starting problems, refer to Engine Starting Fault Isolation Chart (Ref. Fig. 101).

C. Engine Operating

(1) To diagnose and rectify engine operating problems, refer to Engine Operating Fault Isolation Chart (Ref. Fig. 102).

D. Engine Performance

(1) To diagnose and rectify engine performance problems, refer to Engine Performance Fault Isolation Chart (Ref. Fig. 103).

E. Engine Condition Trend Monitoring Shift

(1) To diagnose and rectify engine condition trend monitoring shift, refer to ECTM Shift Fault Isolation Chart (Ref. Table 101) .

F. Engine Lubrication

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(1) To diagnose and rectify engine lubrication problems, refer to Engine Lubrication Fault Isolation Chart (Ref. Fig. 104).

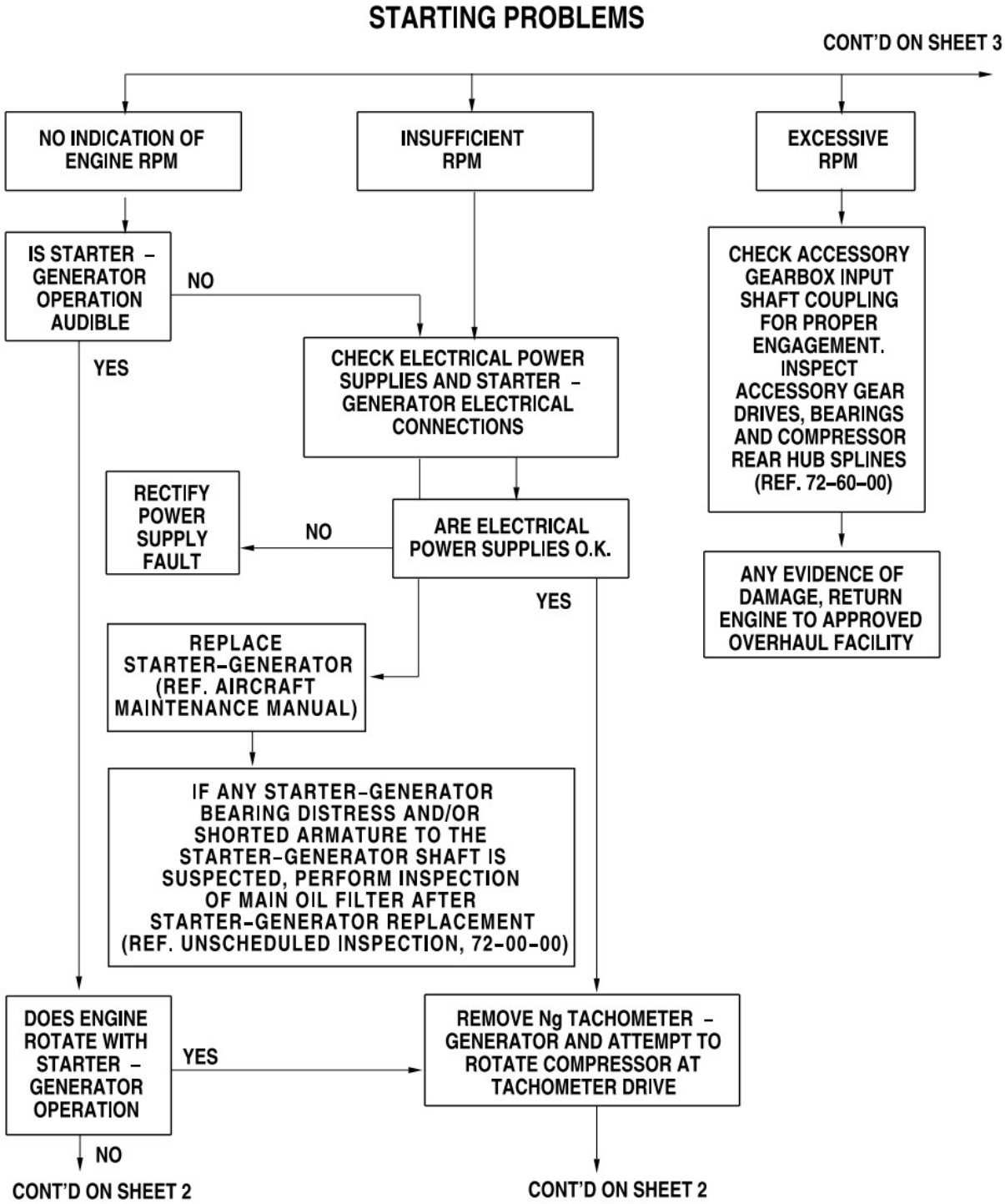
G. Lubricating Oil Contamination

(1) To diagnose and rectify engine lubricating oil contamination, refer to Engine Lubricating Oil Contamination Fault Isolation Chart (Ref. Fig. 105).

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Figure 101 Engine Starting Fault Isolation Chart

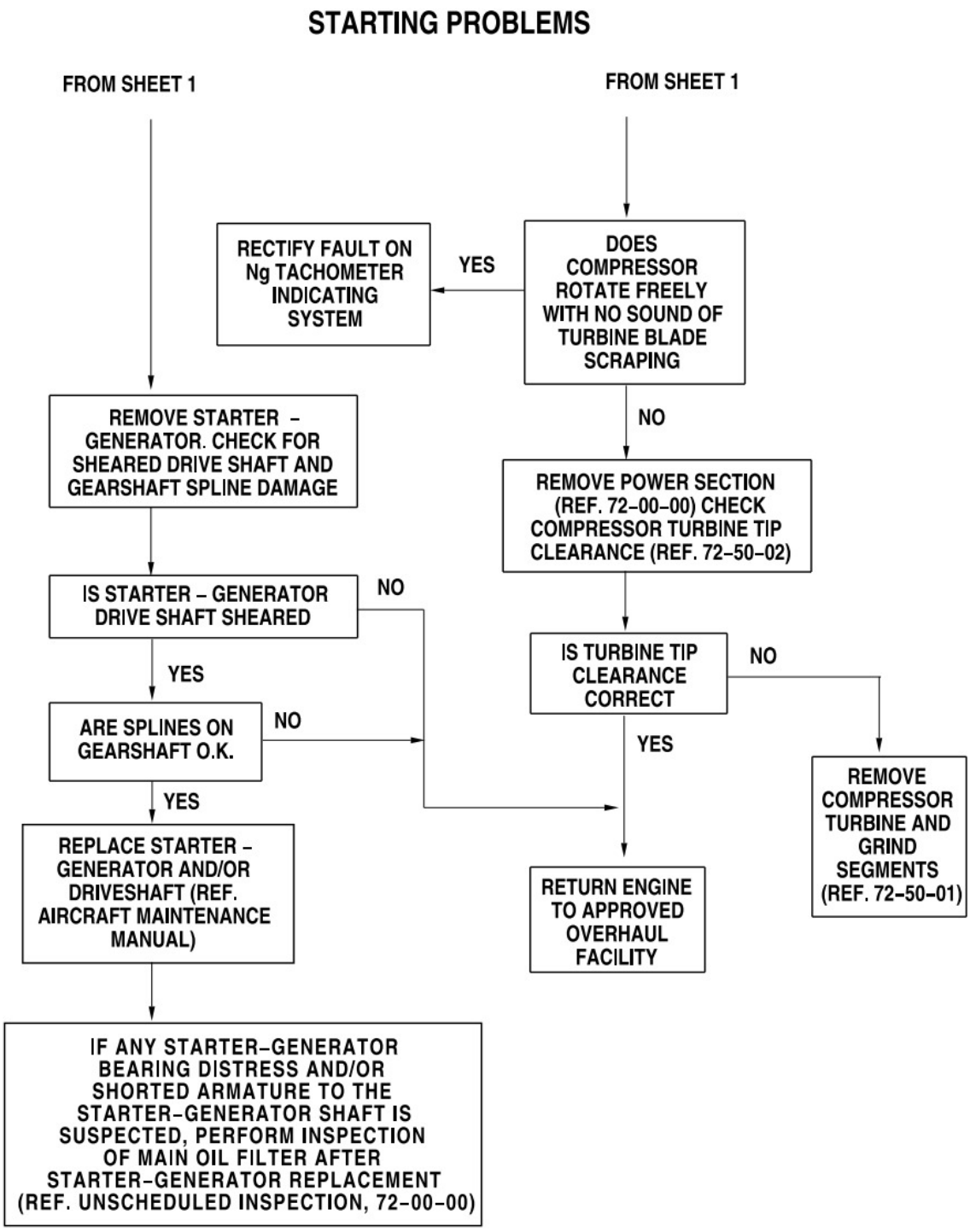
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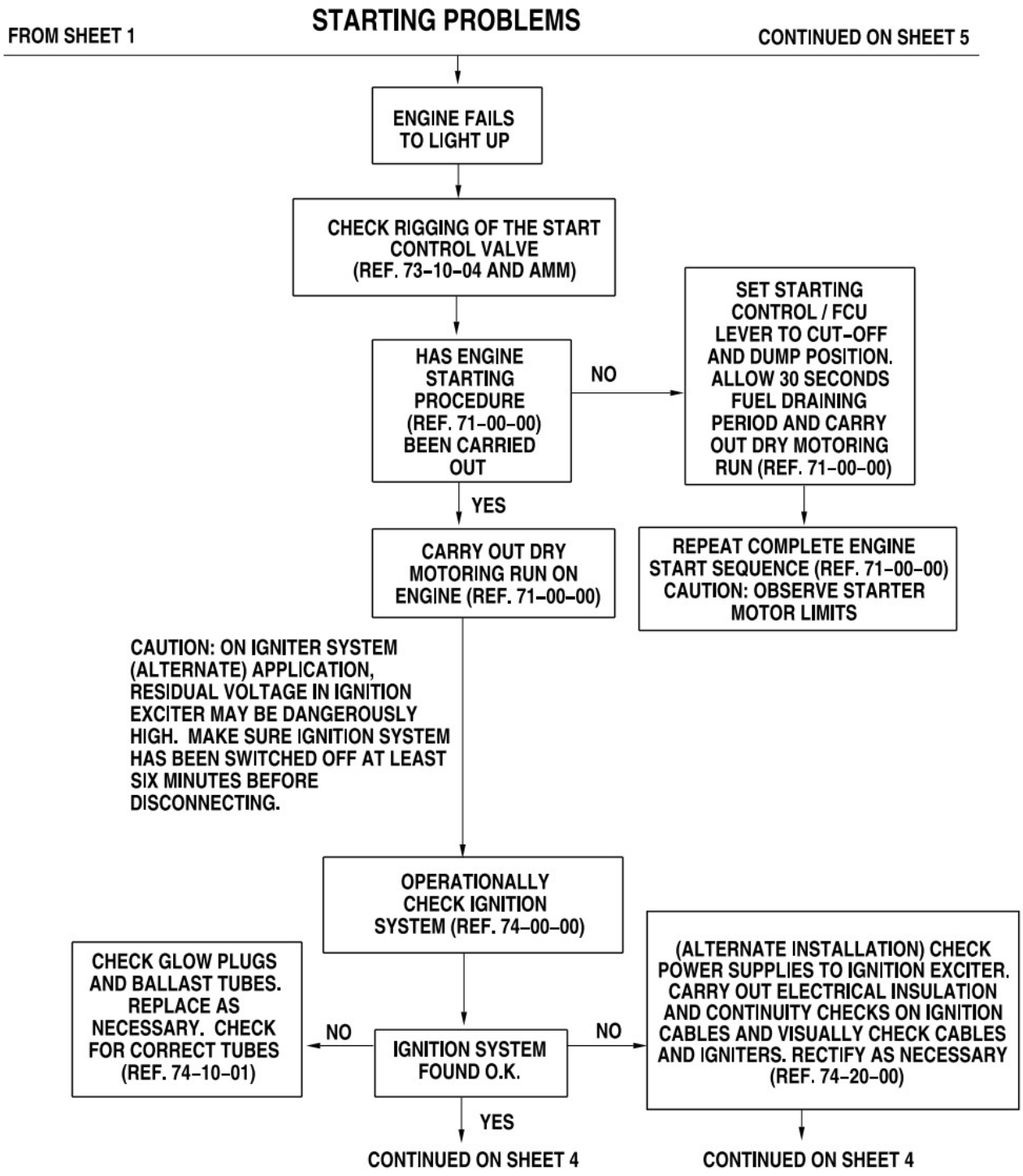
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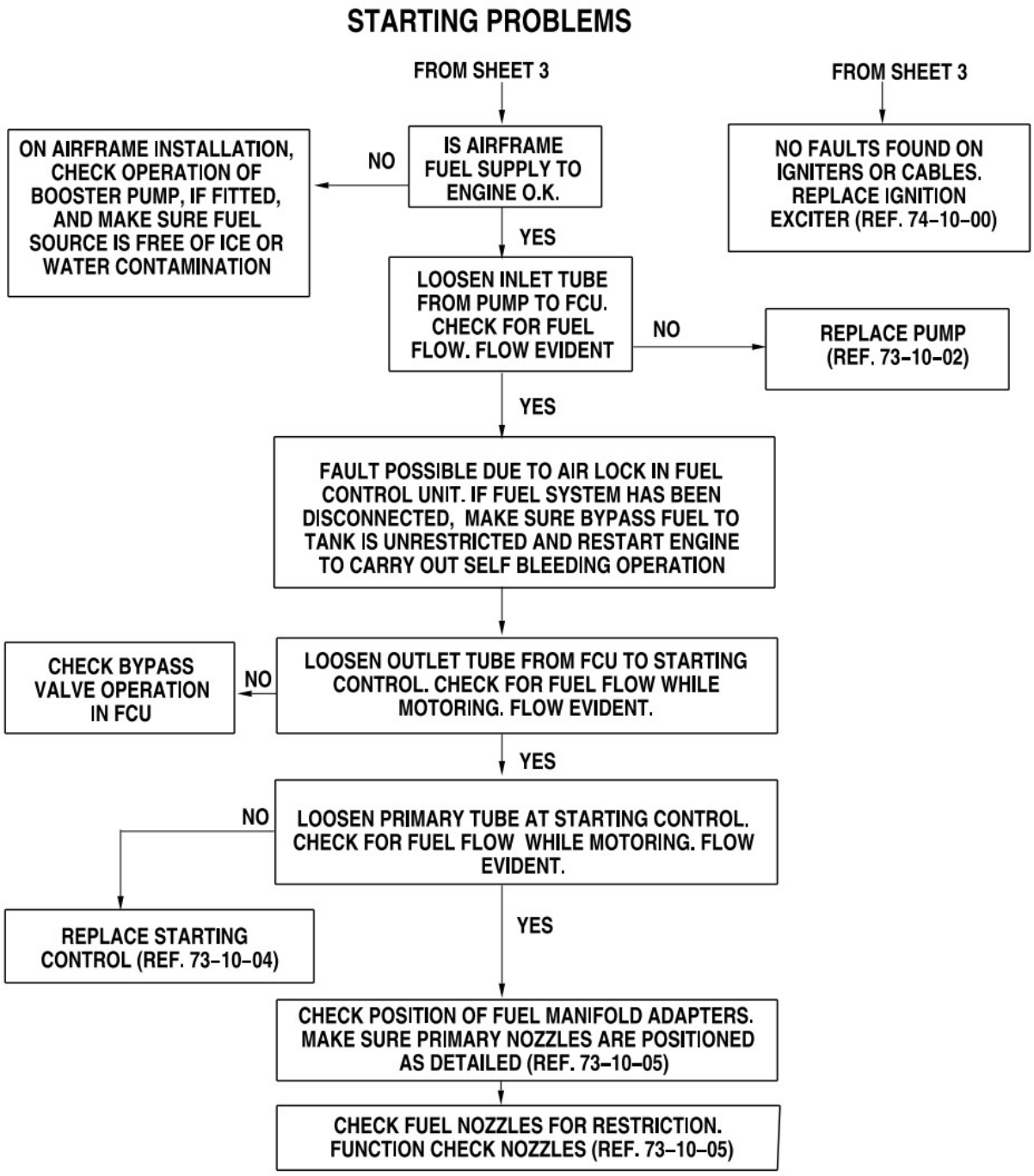
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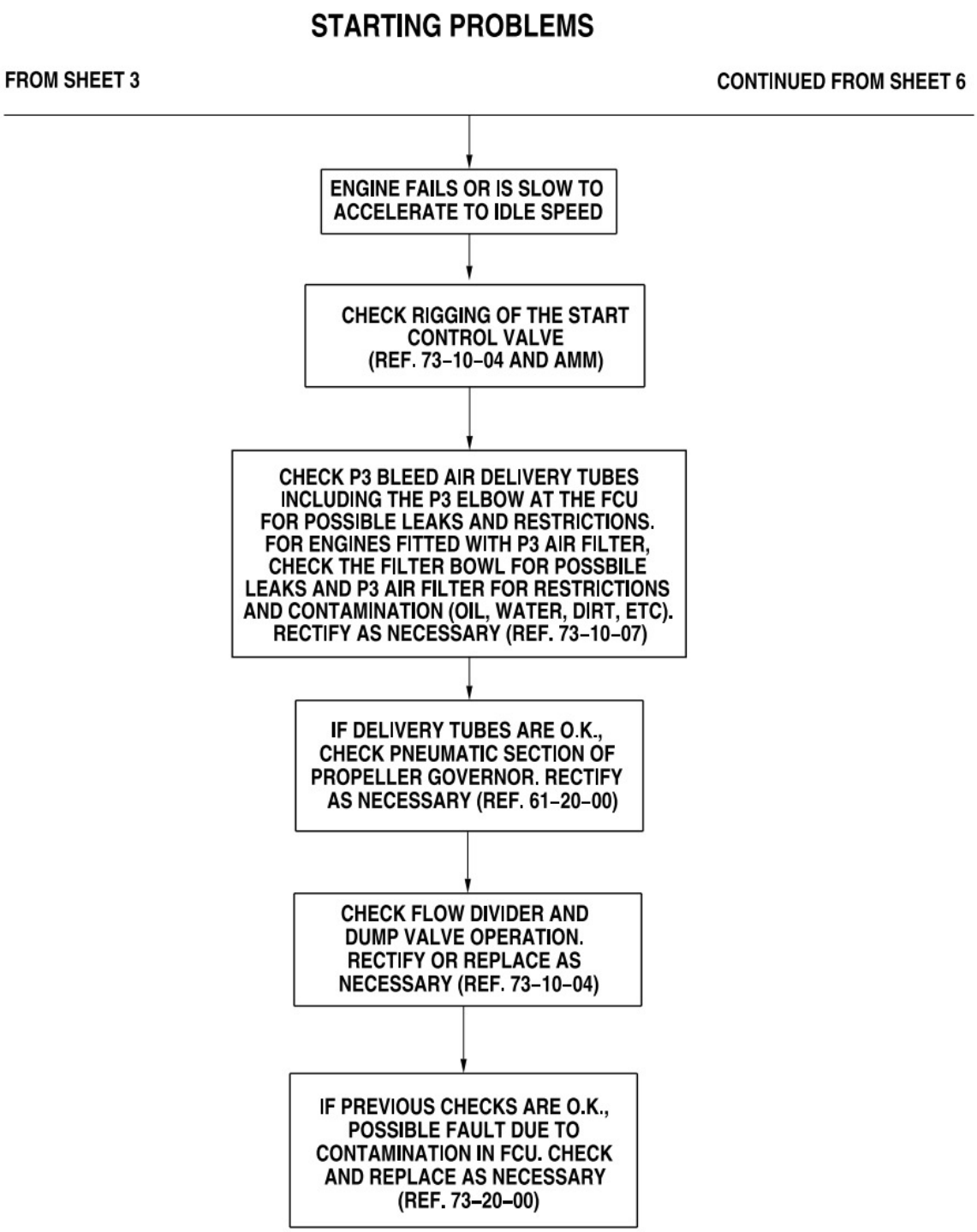
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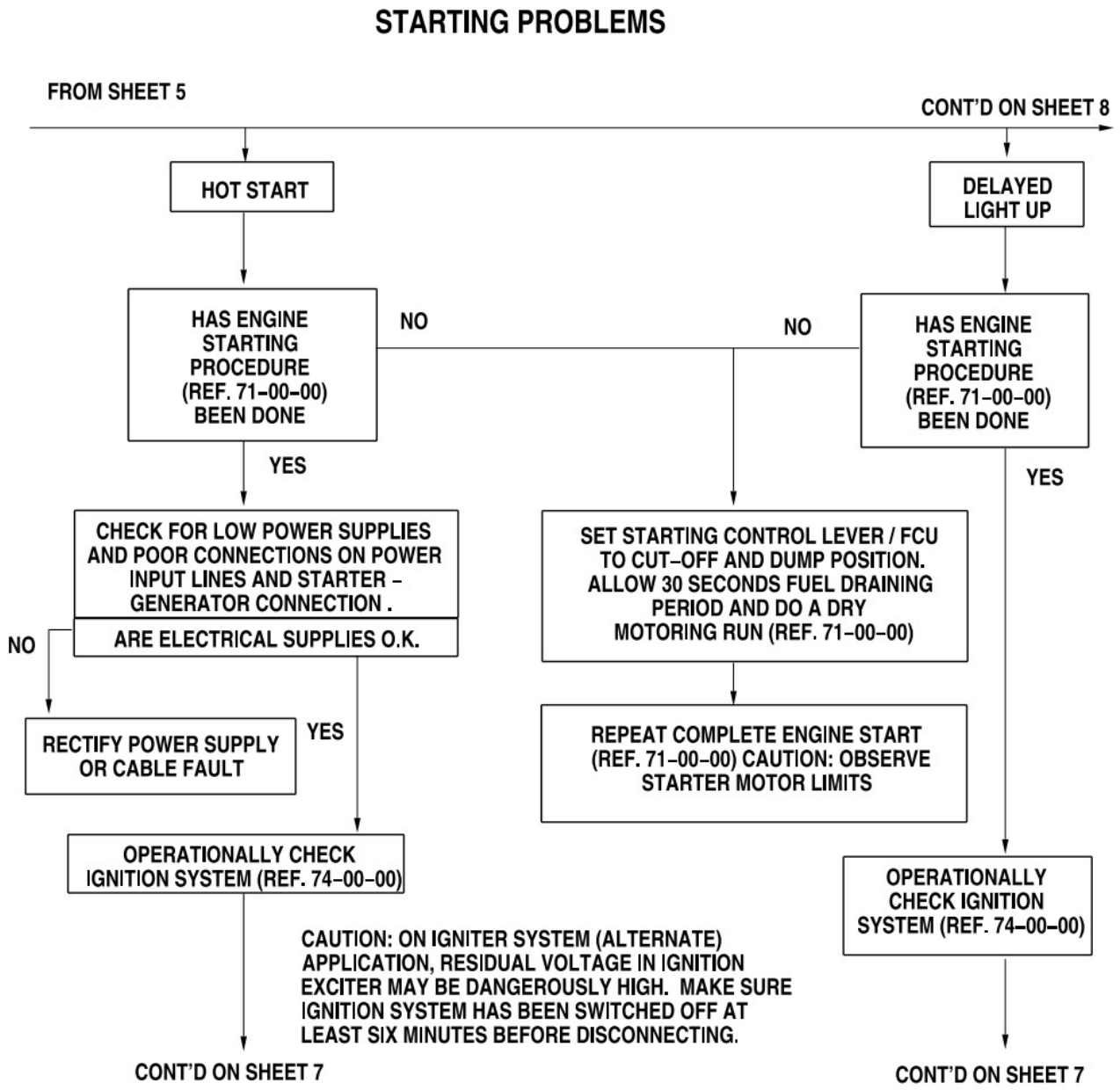
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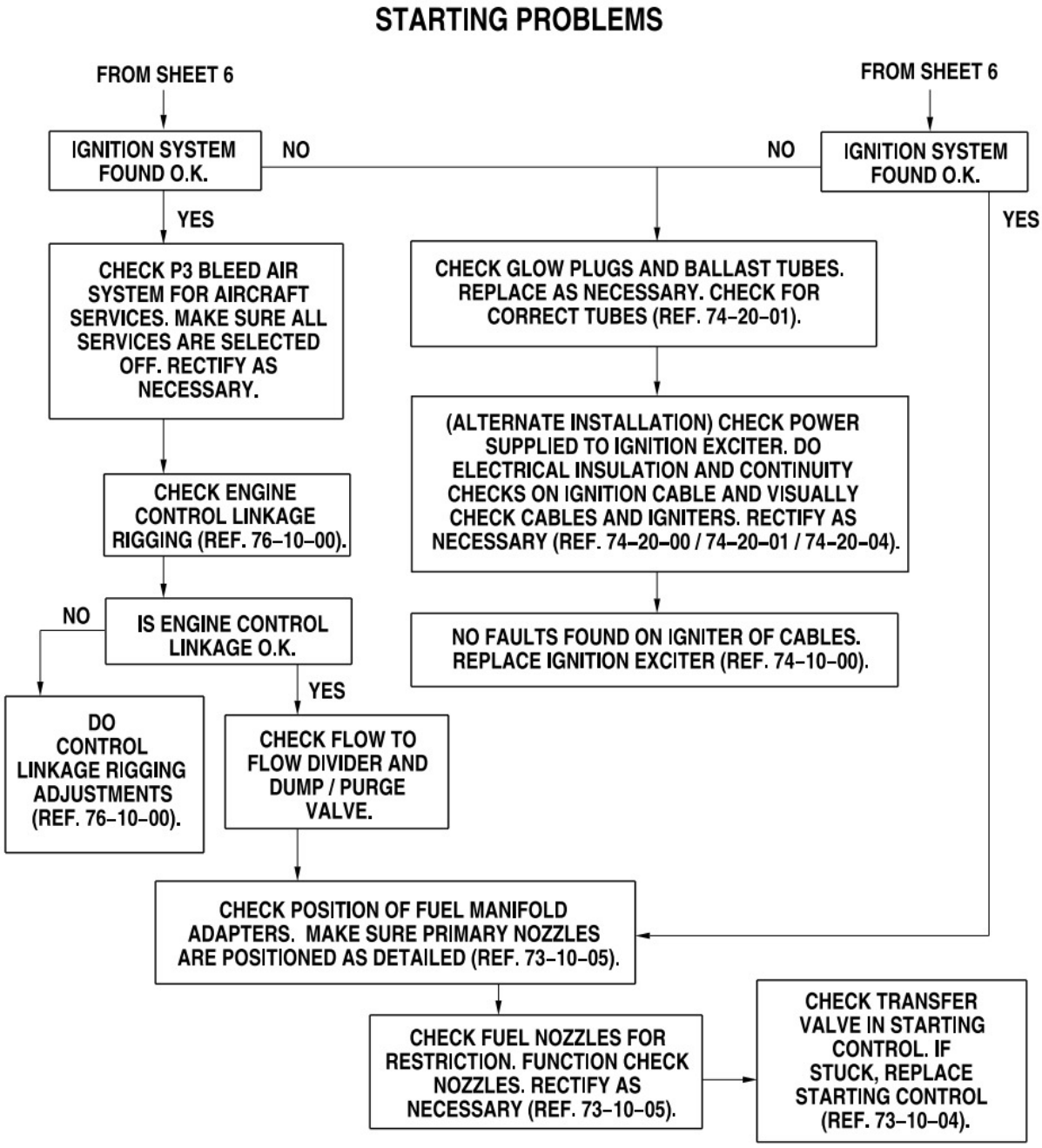
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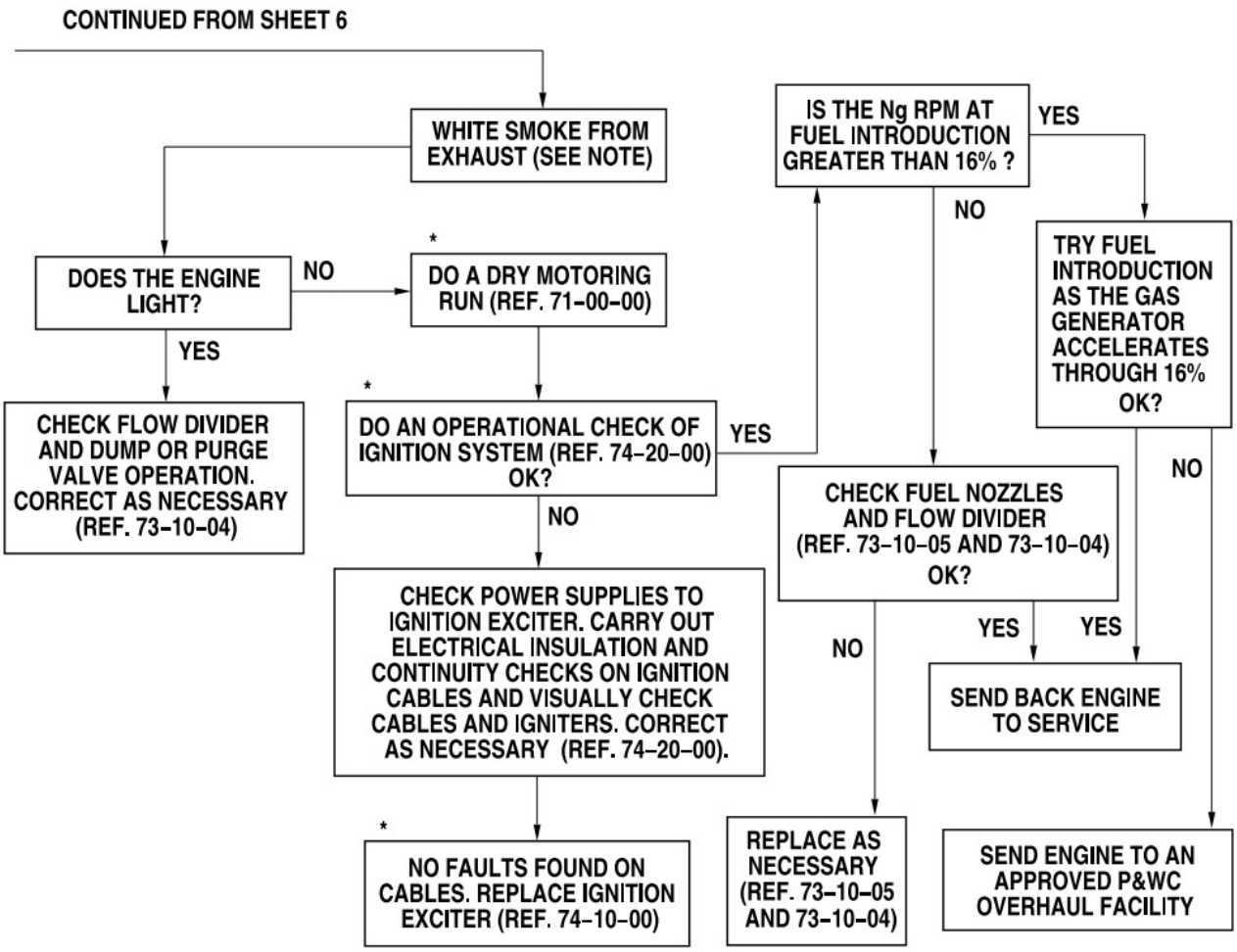
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*** CAUTION:**

1. **RESIDUAL VOLTAGE IN IGNITION EXCITER MAY BE DANGEROUSLY HIGH. MAKE SURE THAT THE IGNITION SYSTEM HAS BEEN SWITCHED OFF AT LEAST SIX MINUTES BEFORE YOU DISCONNECT.**
2. **MONITOR STARTER MOTOR LIMITS. REFER TO STARTER MOTOR MANUFACTURER'S MANUAL.**

NOTE:

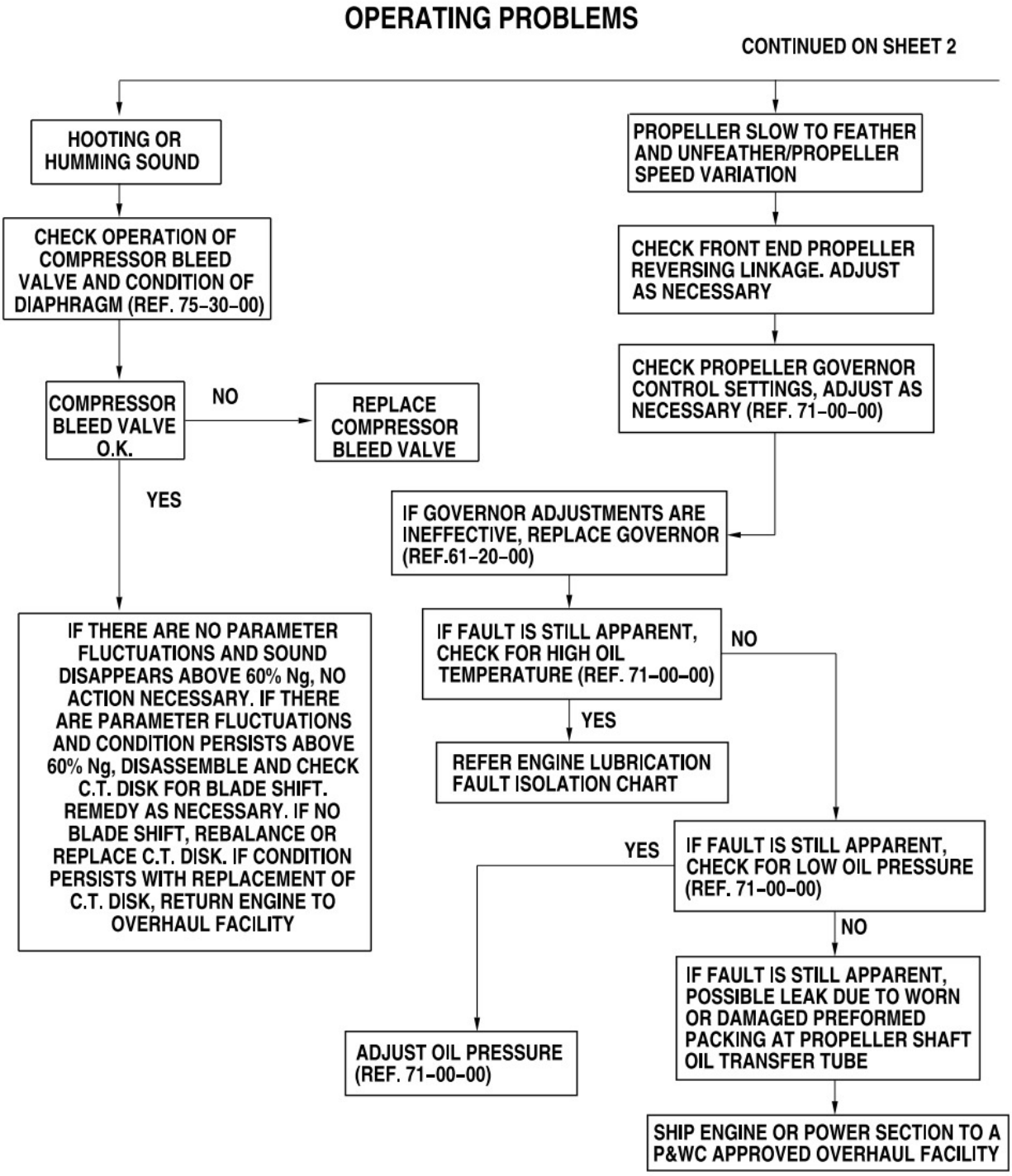
1. **WHITE SMOKE FROM THE EXHAUST IS AN INDICATION THAT FUEL IS IN THE ENGINE AND BECAME A GAS ON START.**
2. **SMOKY STARTS CAN BE PREVENTED BY REPLACEMENT OF FLOW DIVIDER AND PURGE VALVE (REF. 73–10–04).**

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Figure 102 Engine Operating Fault Isolation Chart

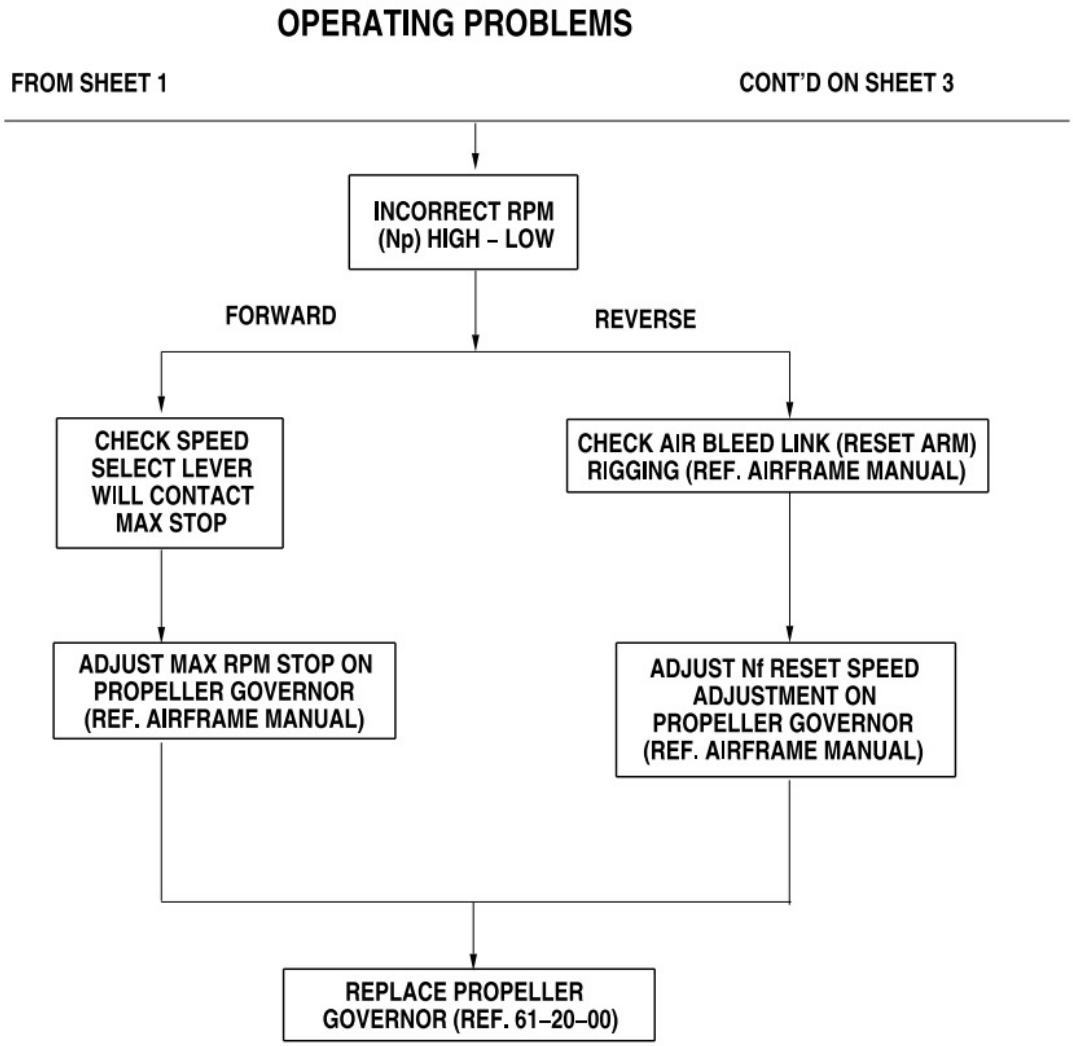
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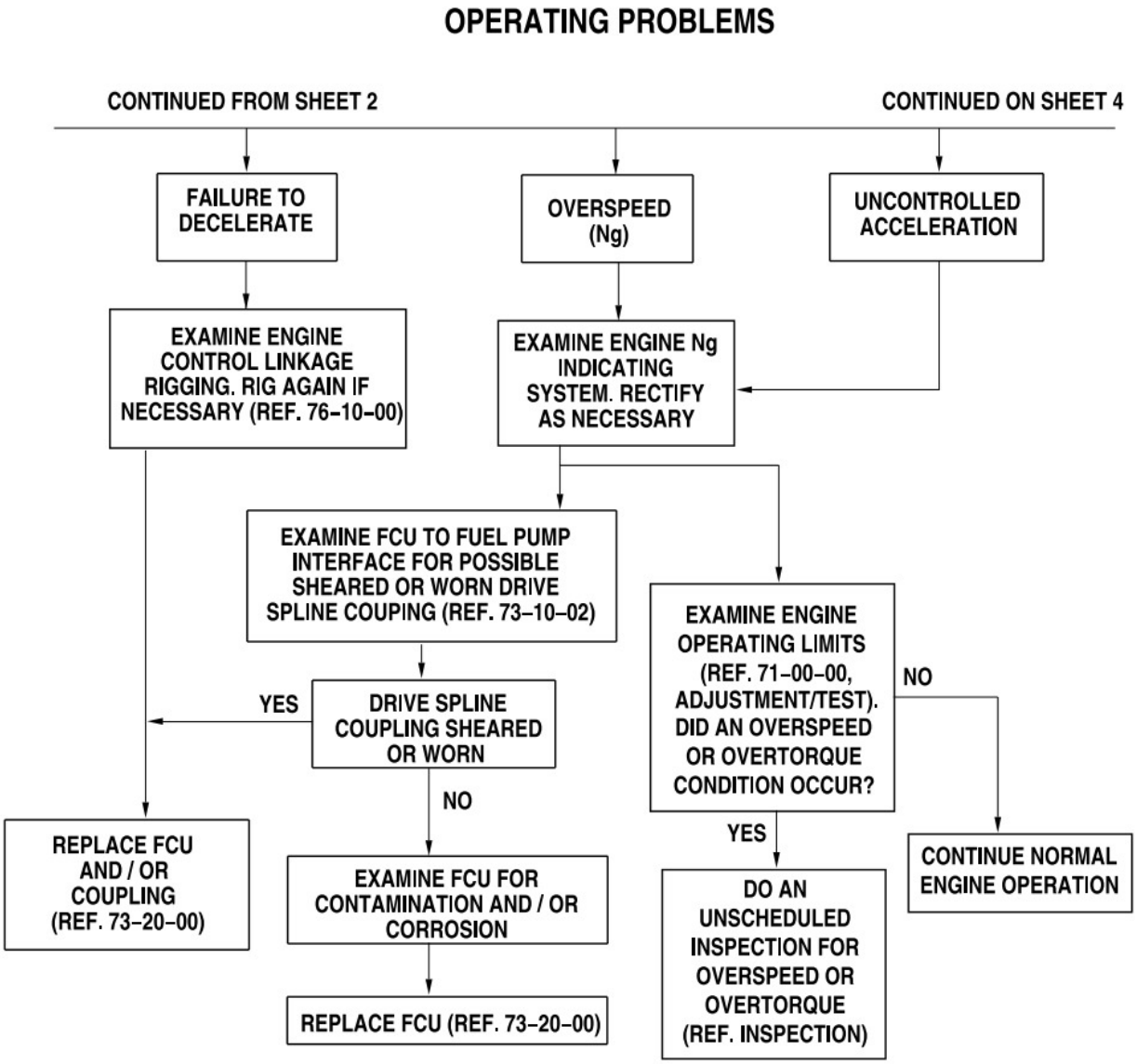
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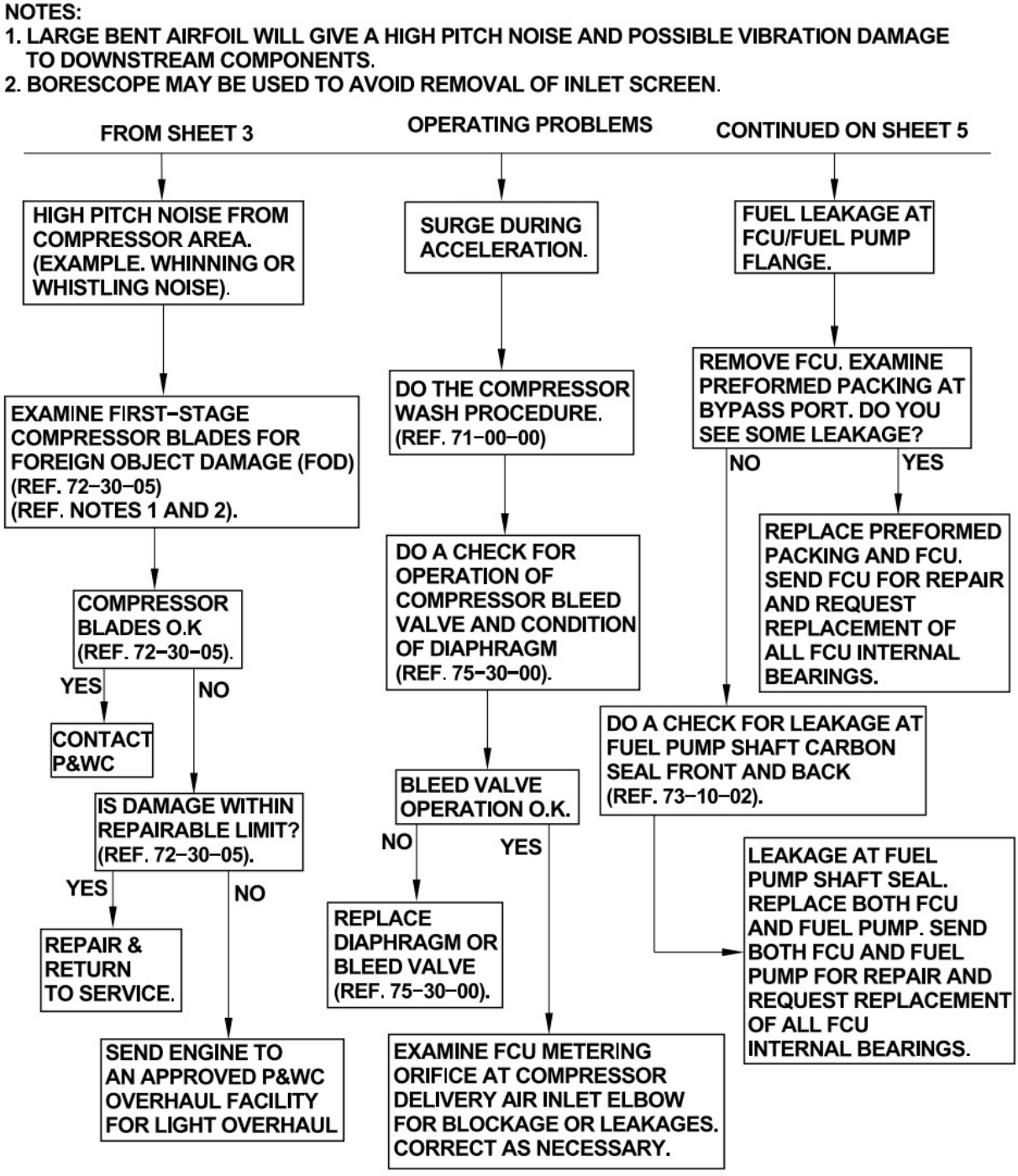
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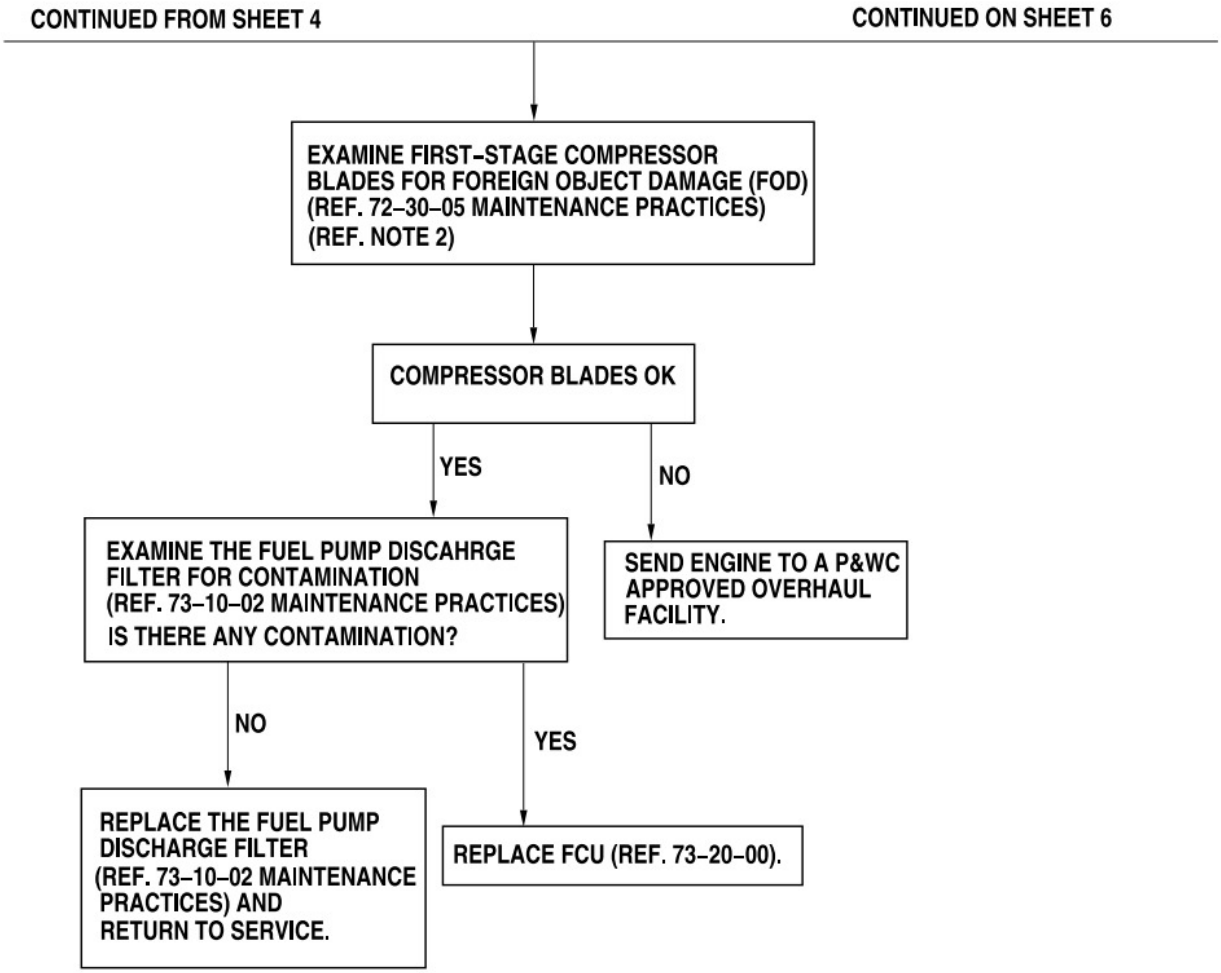
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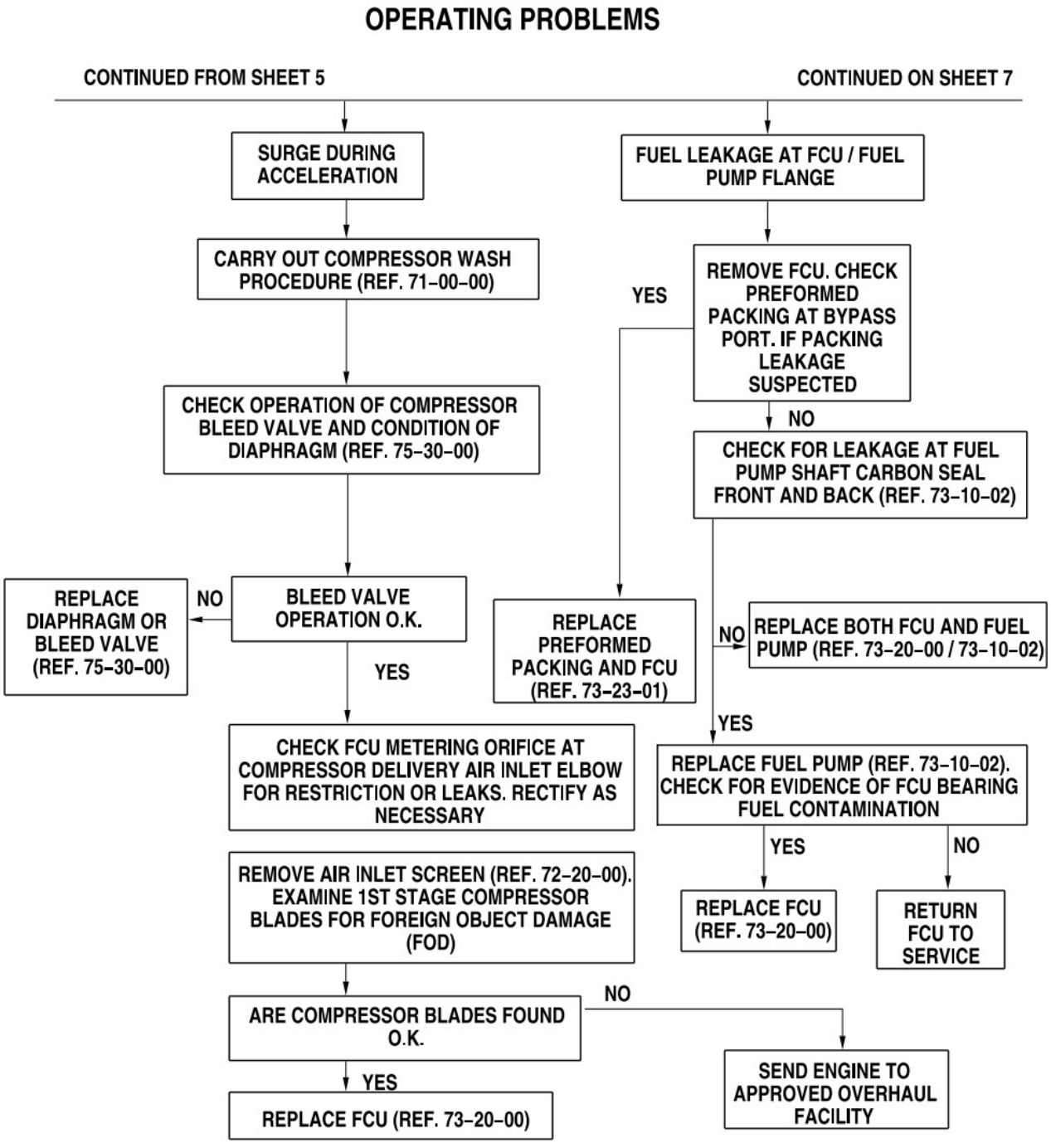
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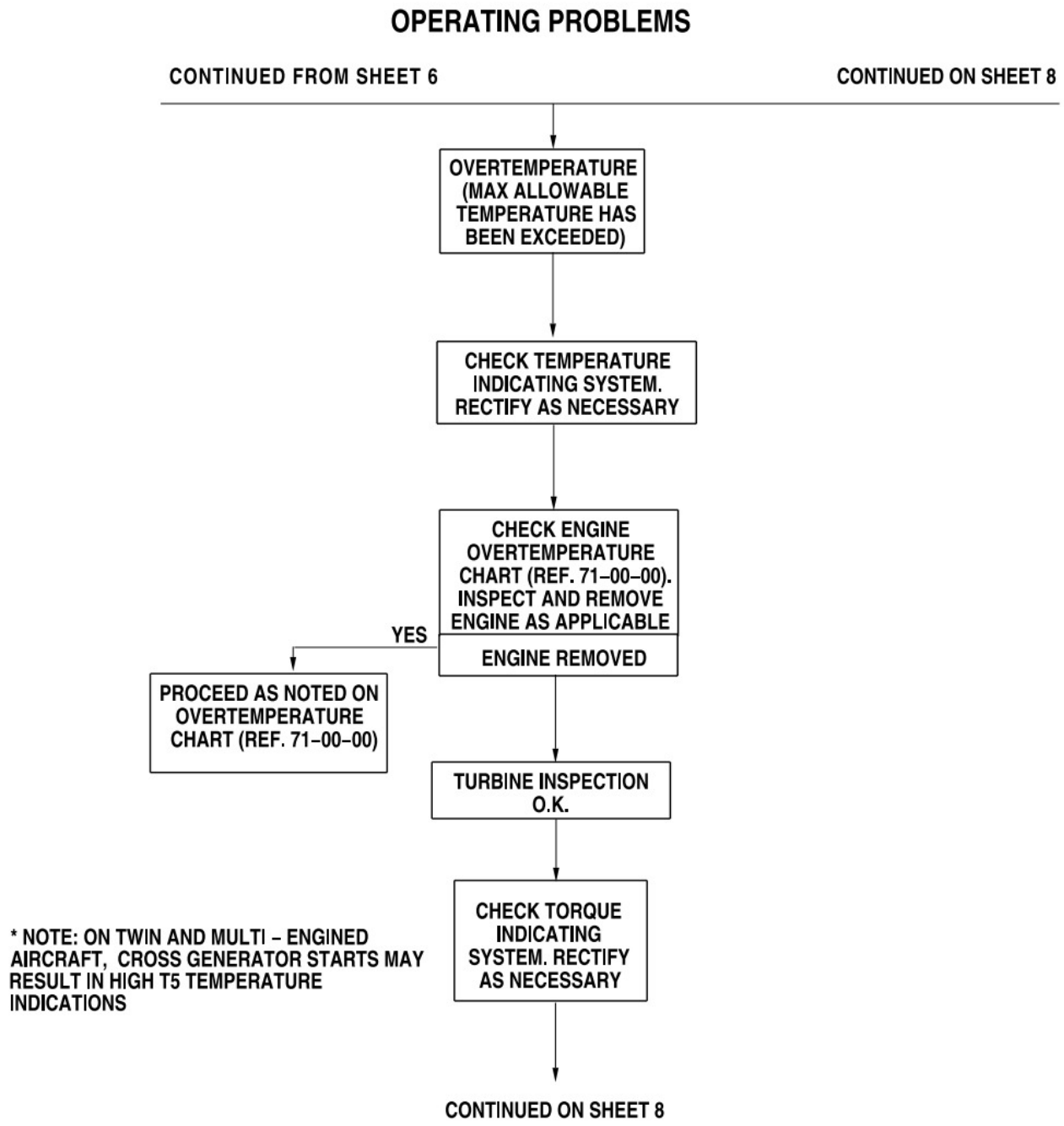
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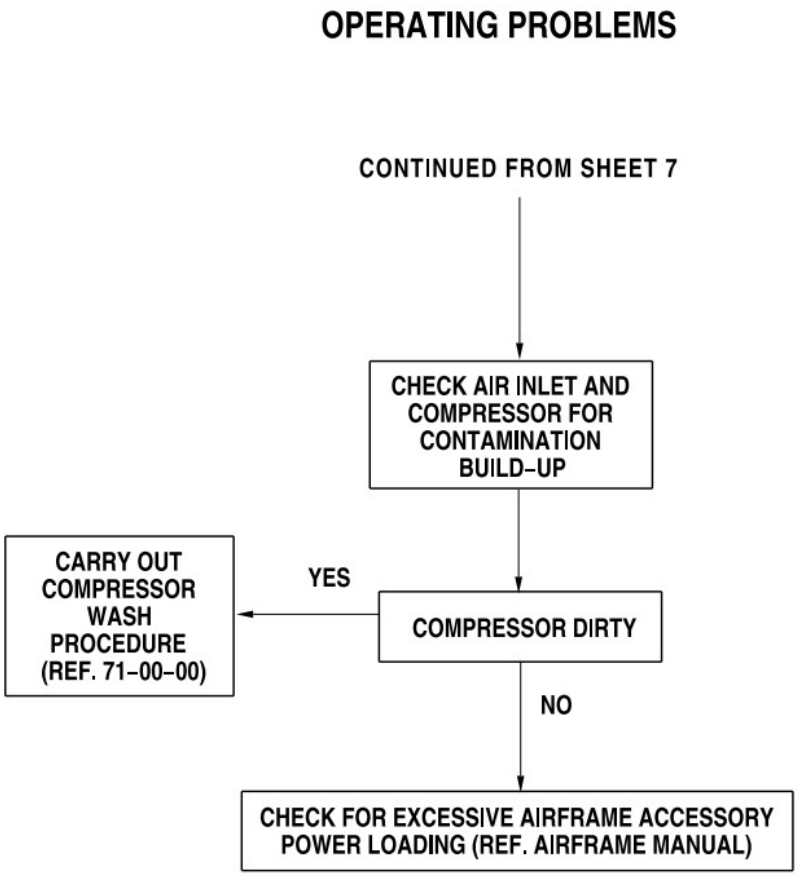
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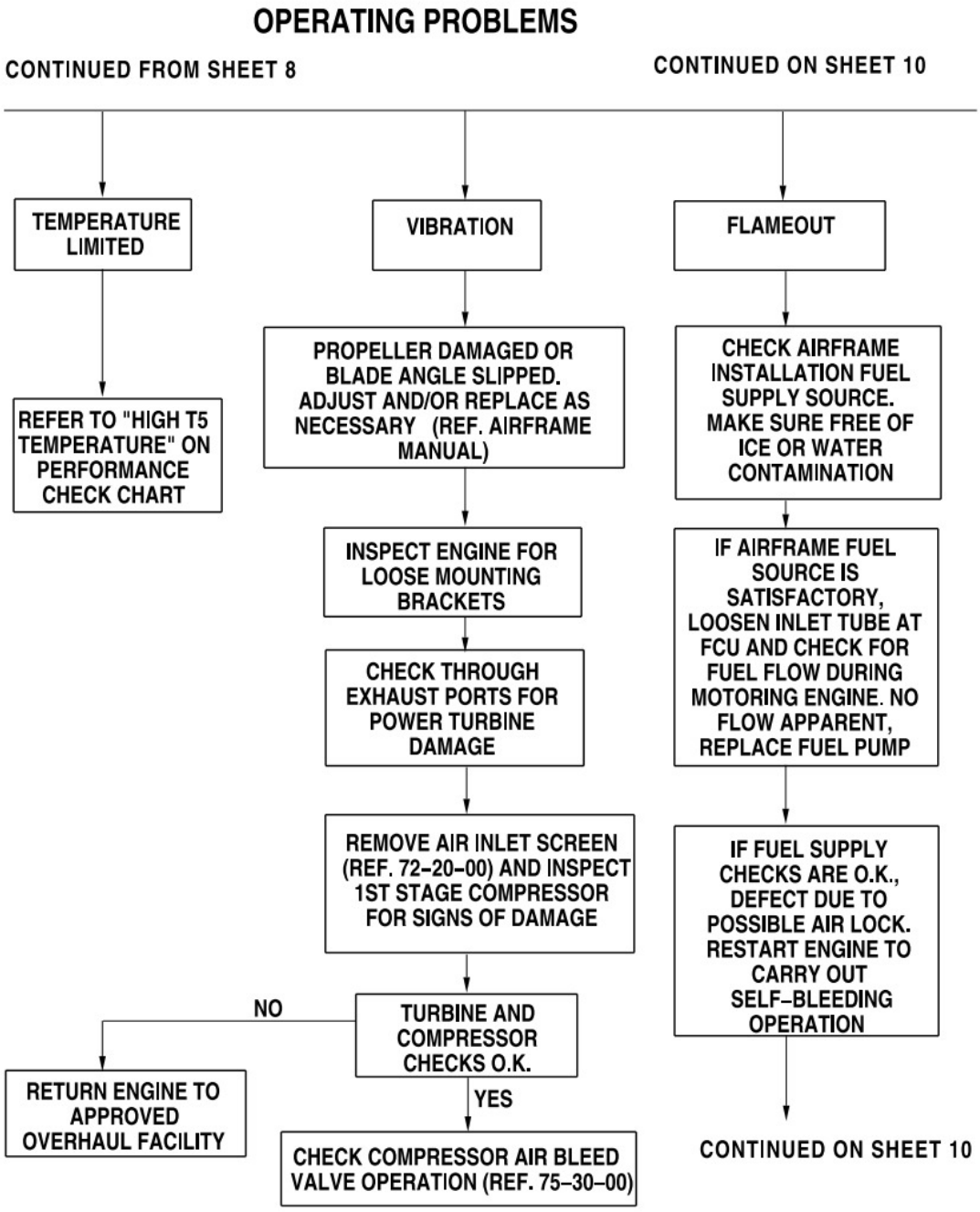
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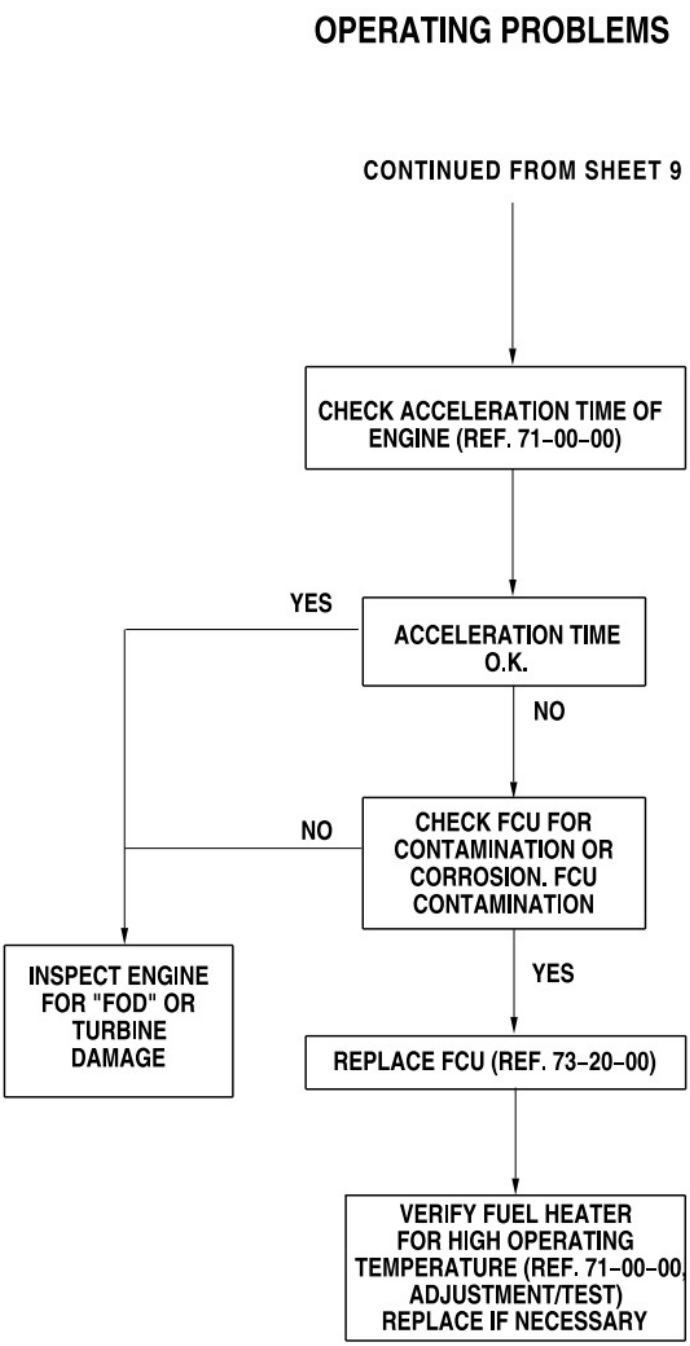
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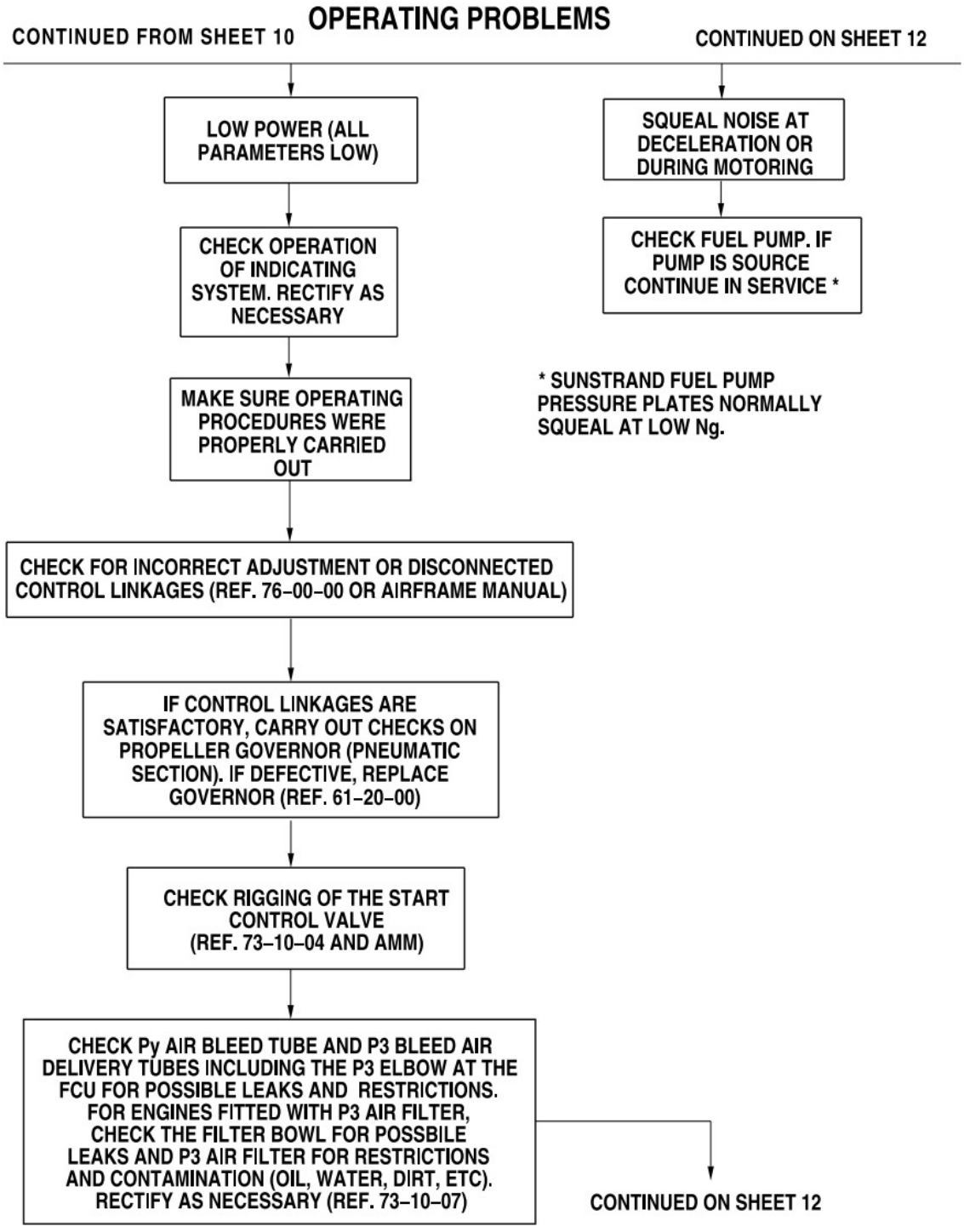
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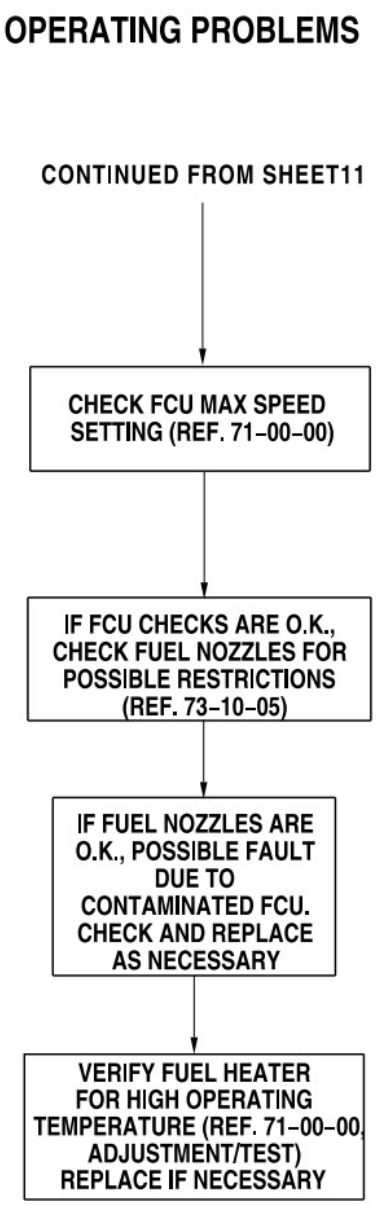
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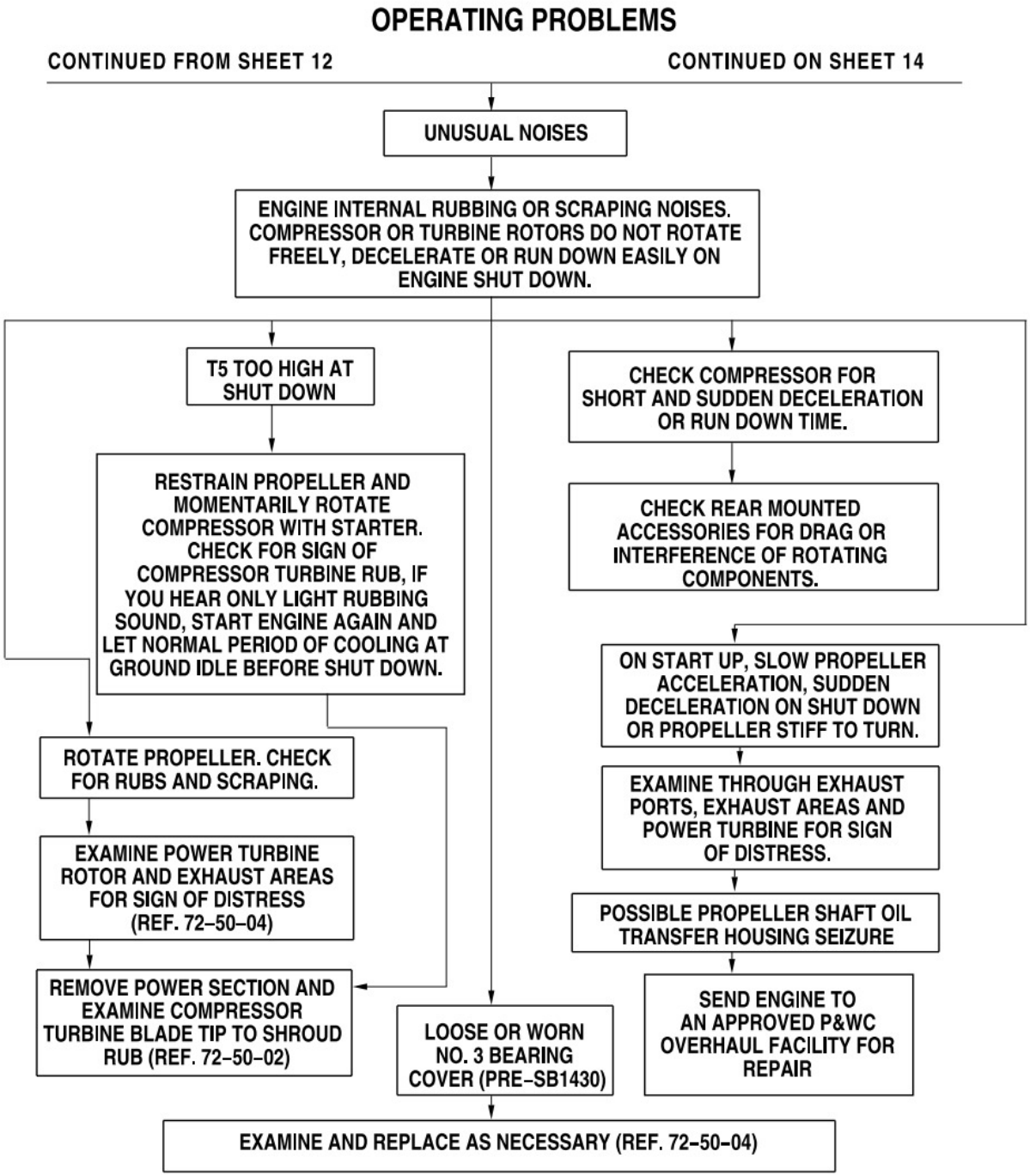
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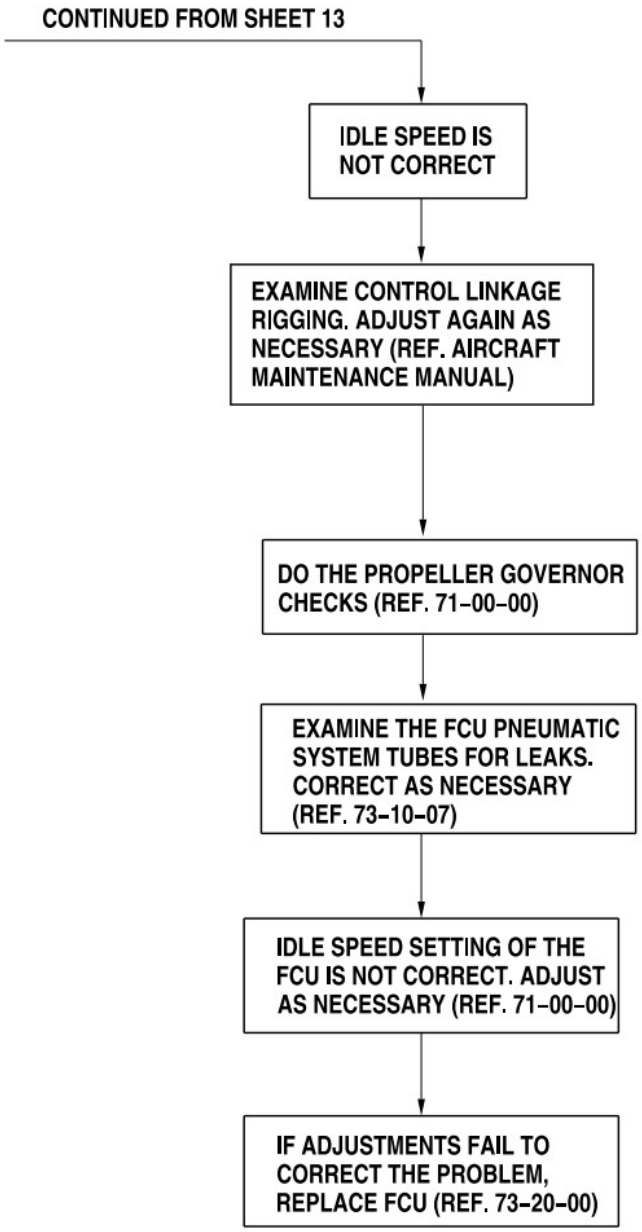
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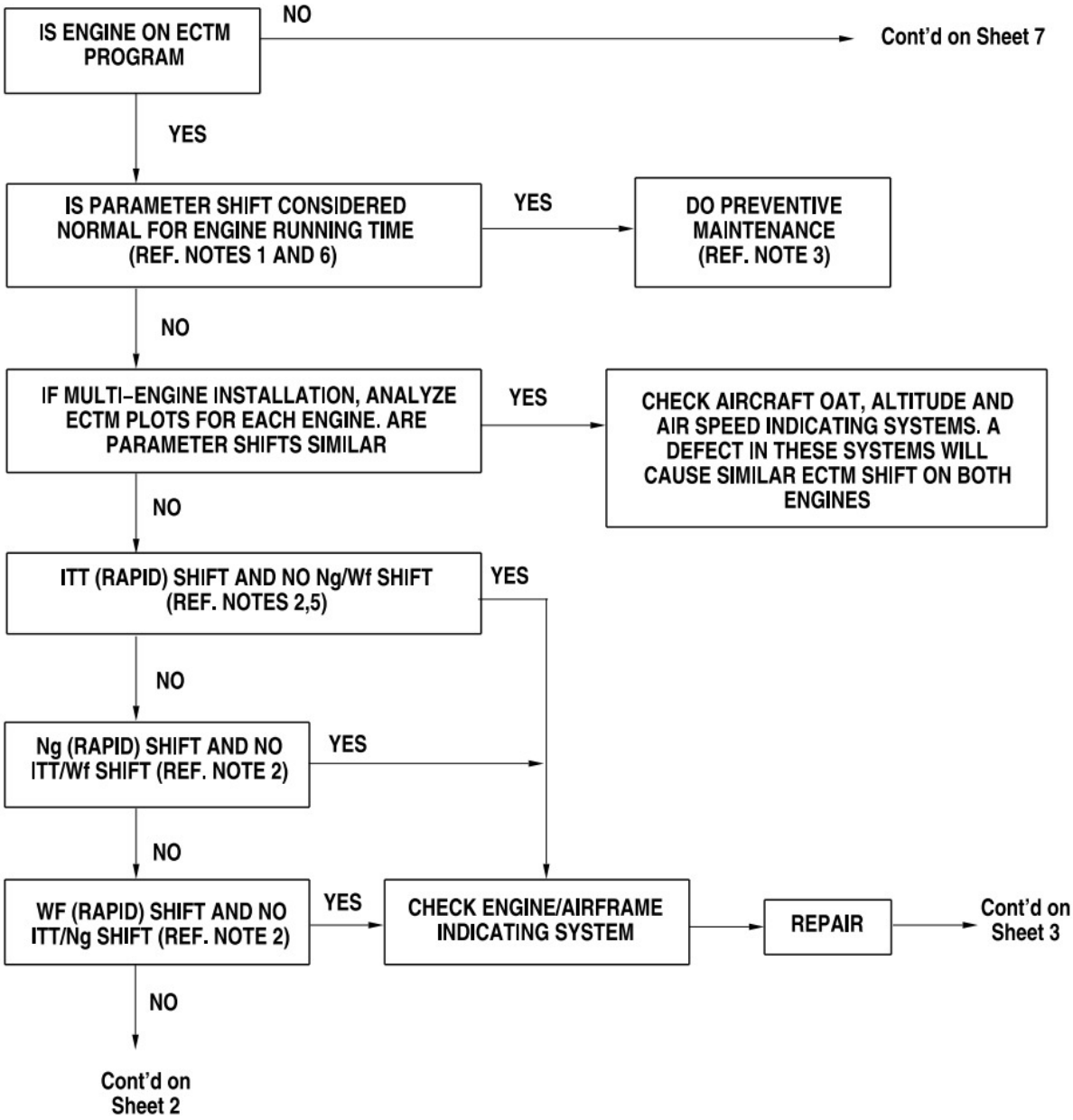


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Figure 103 Engine Performance Fault Isolation Chart

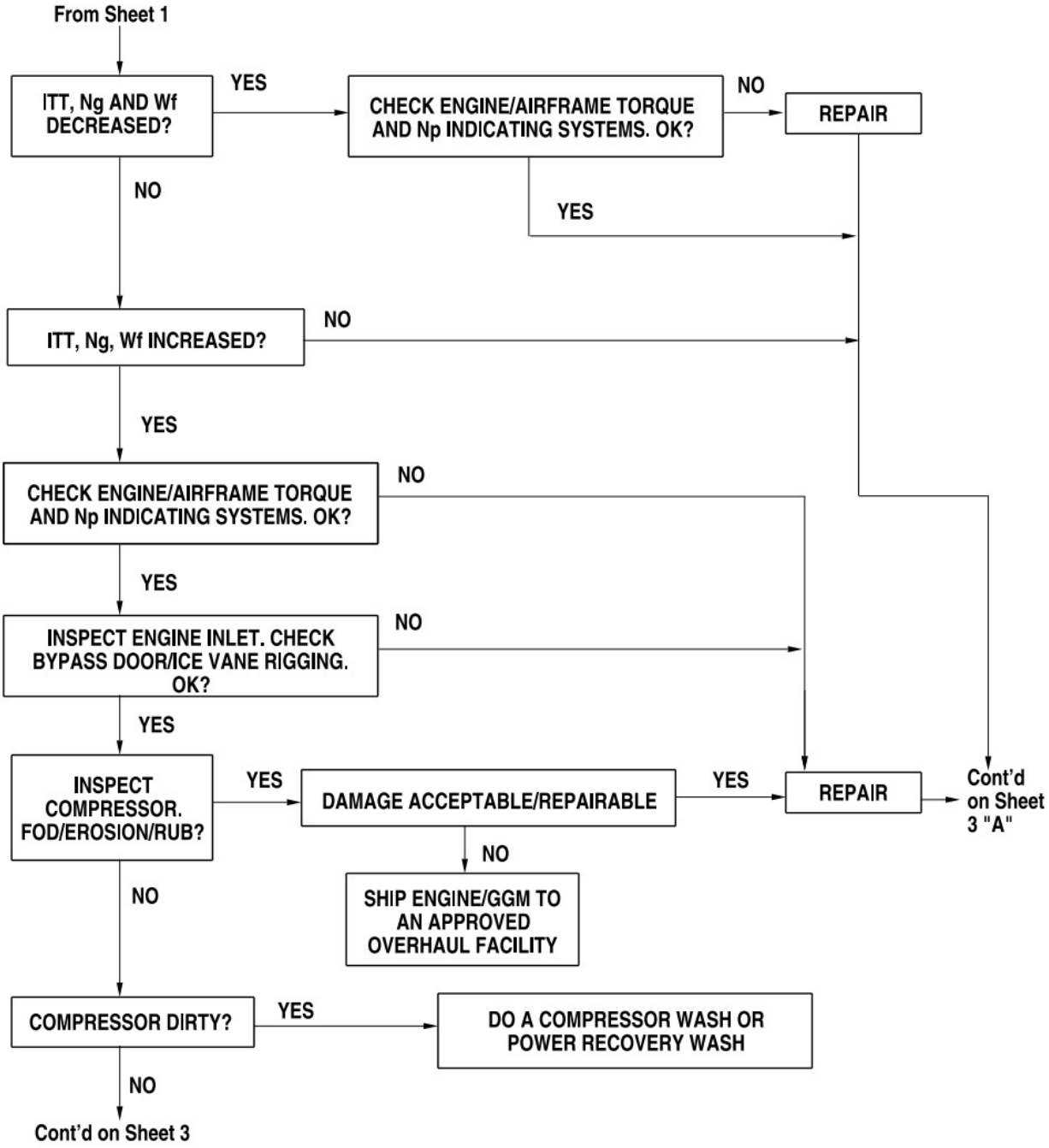
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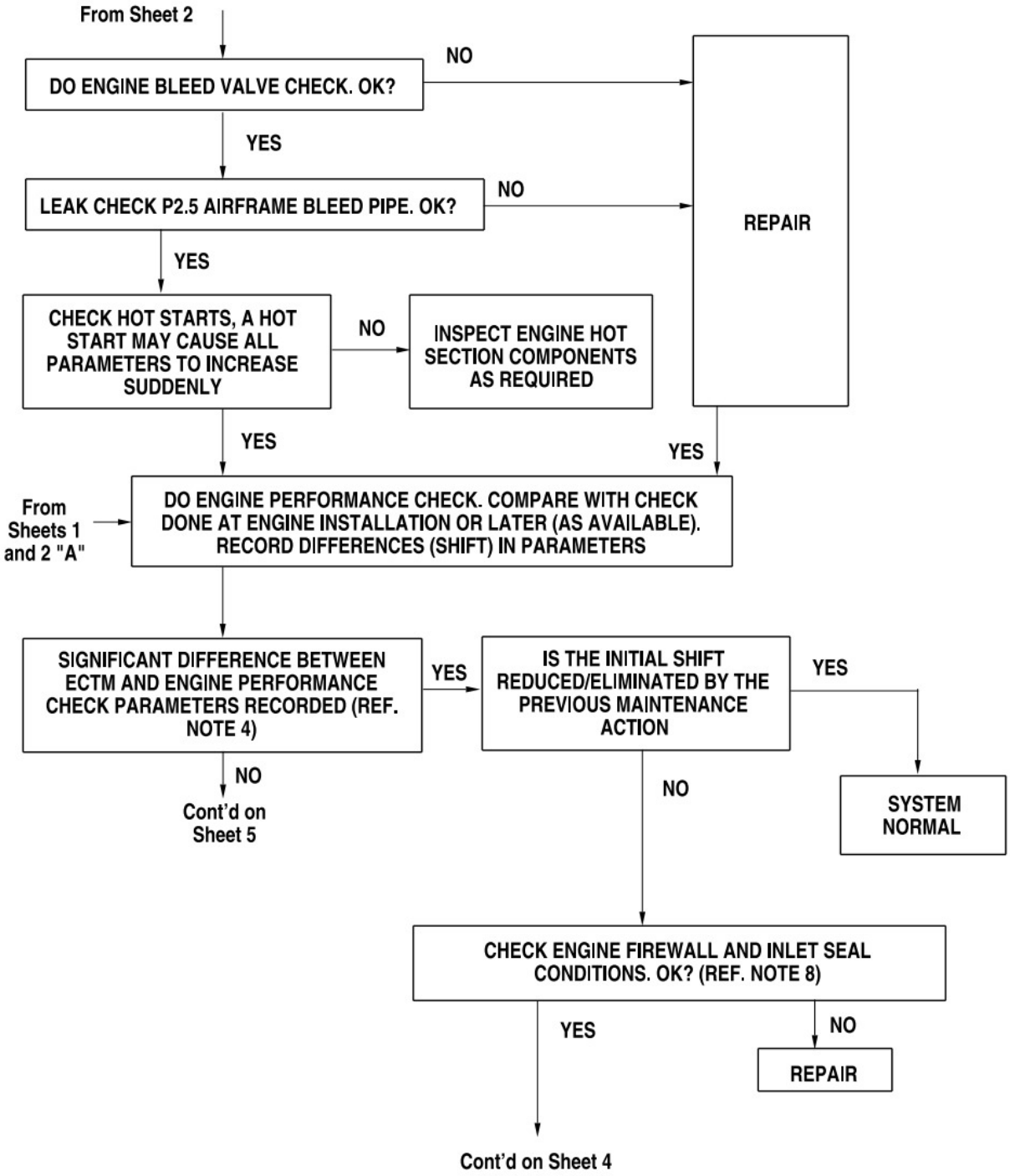
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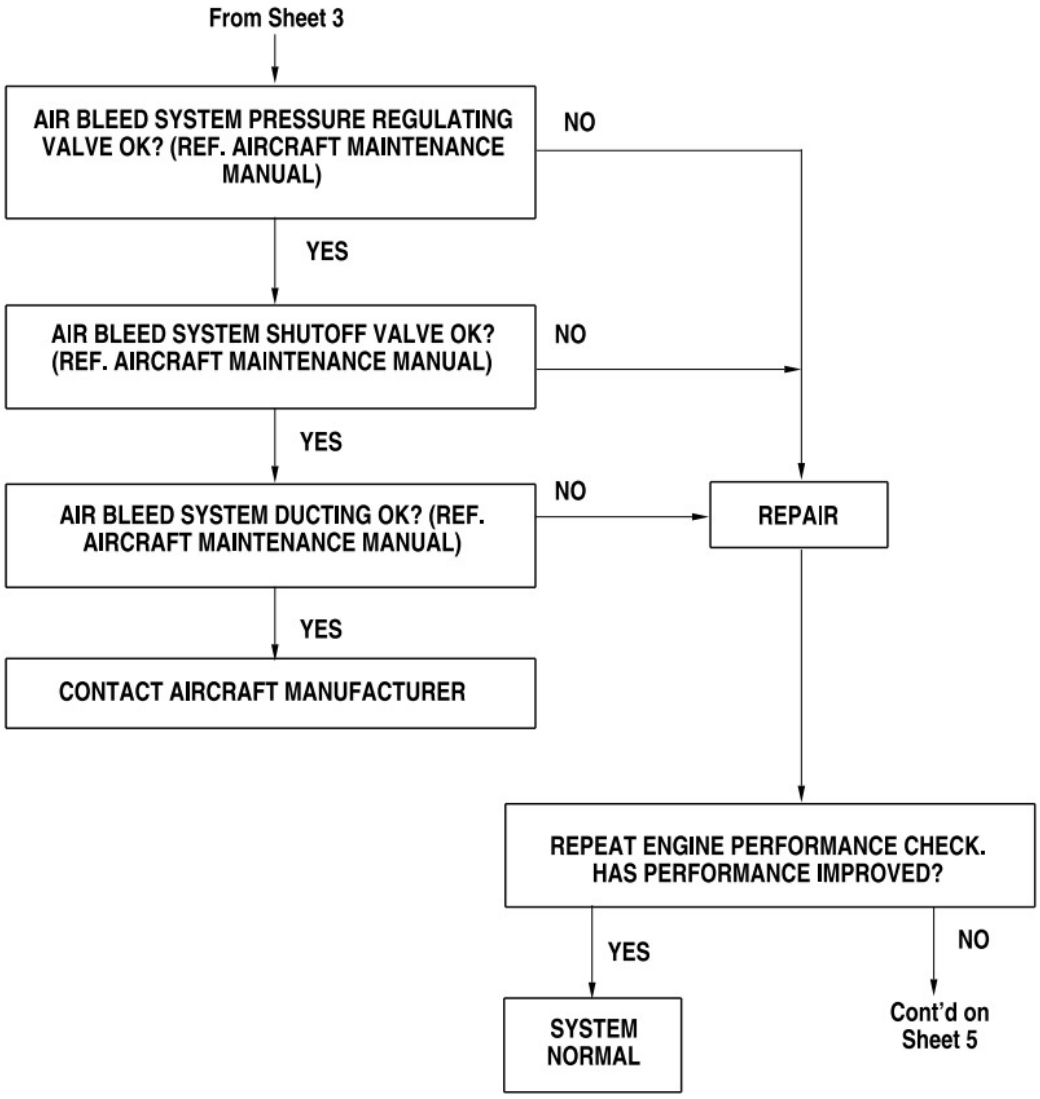
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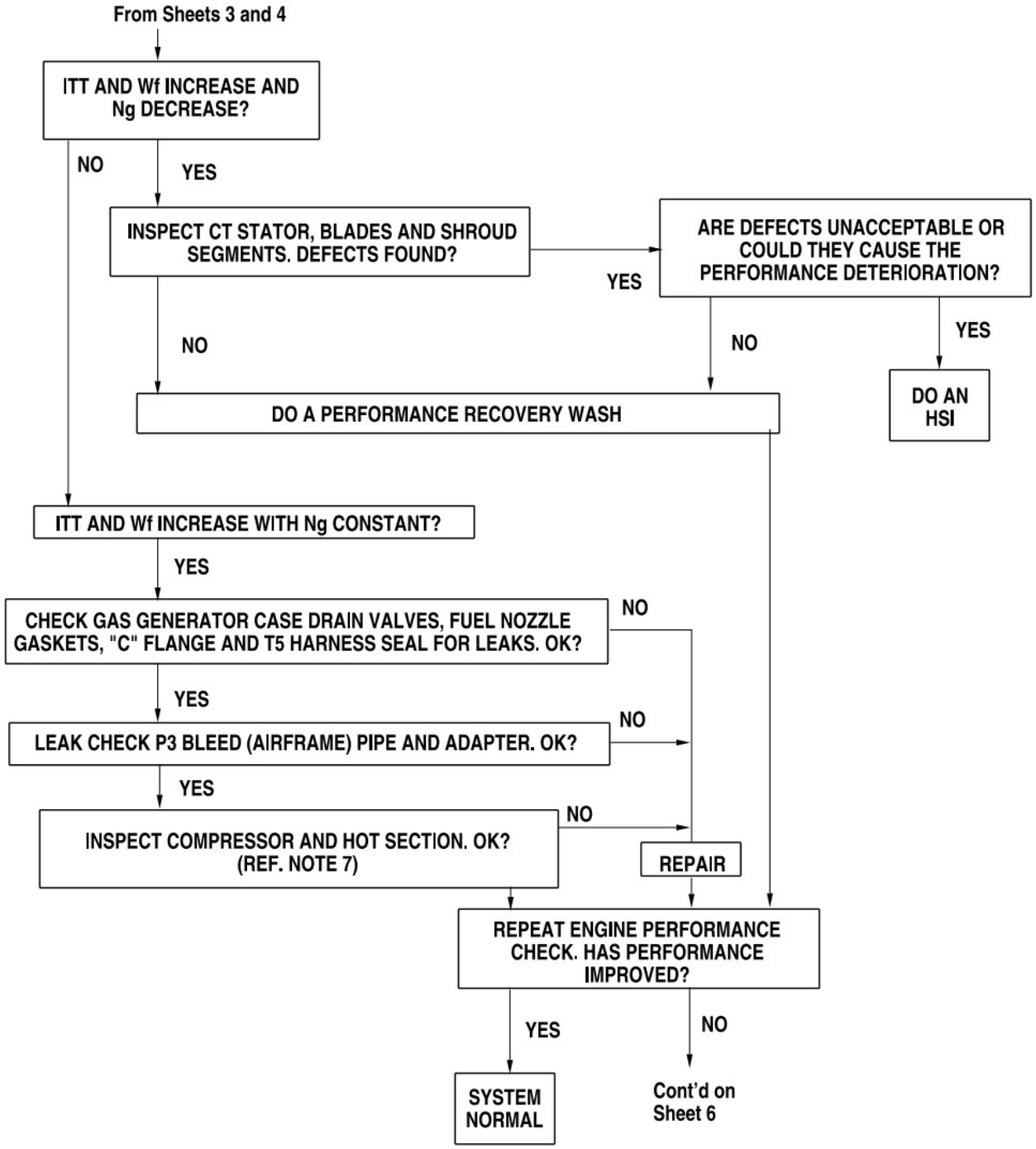
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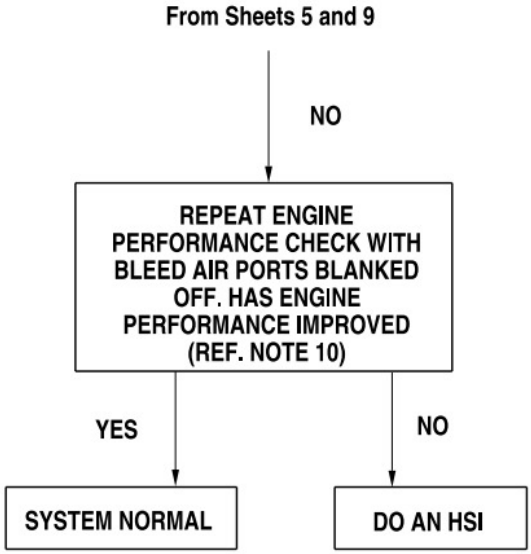
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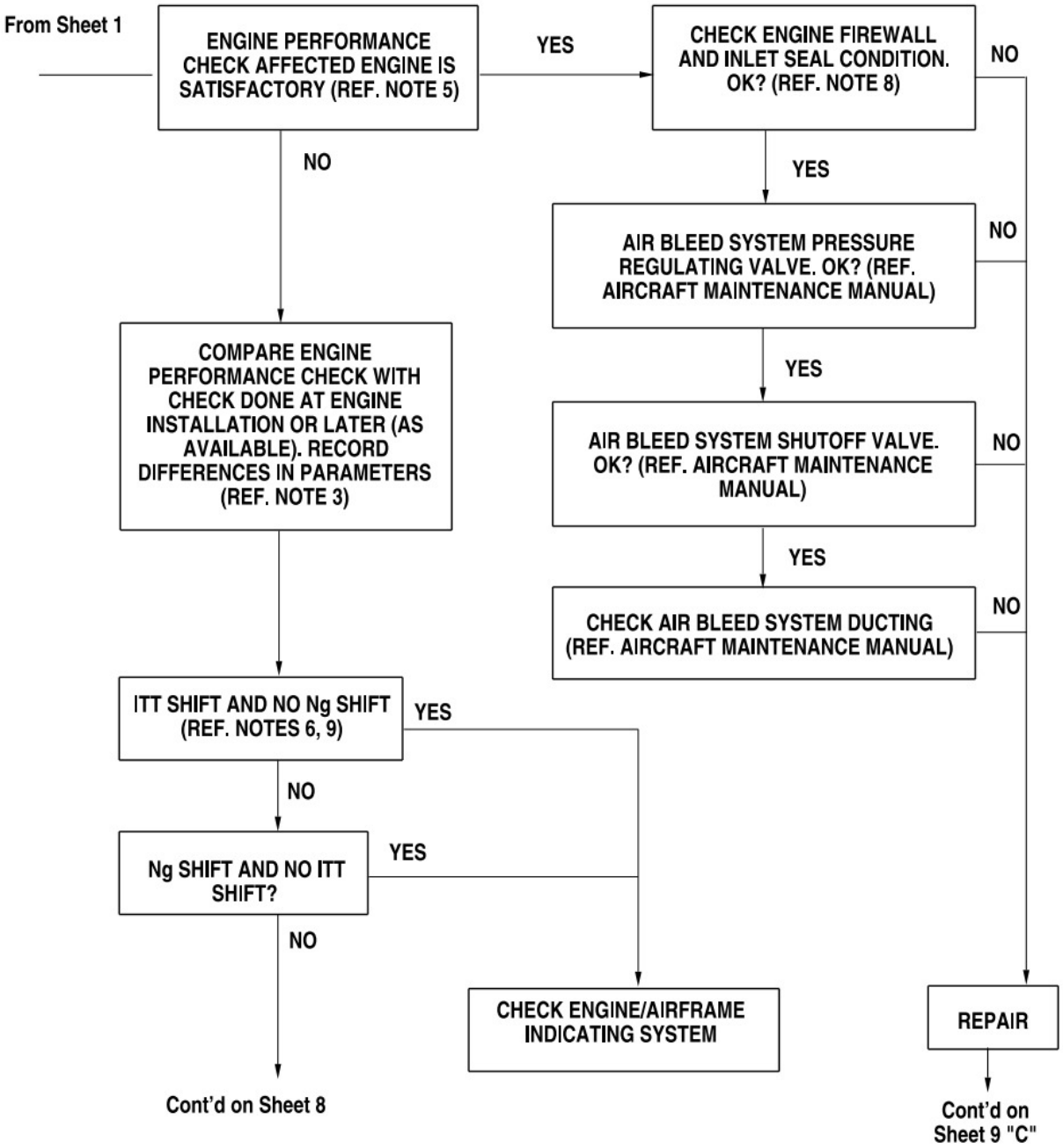
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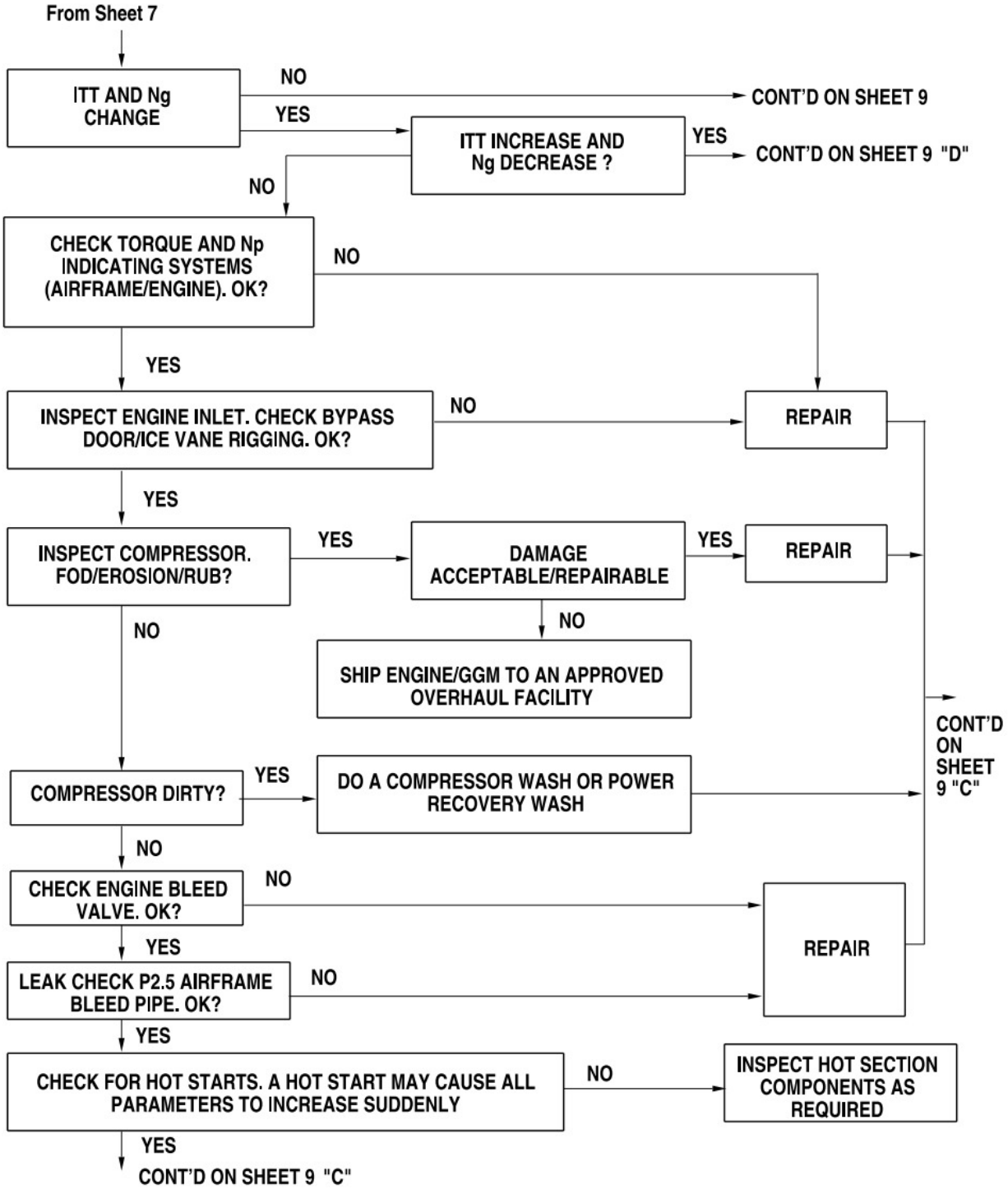
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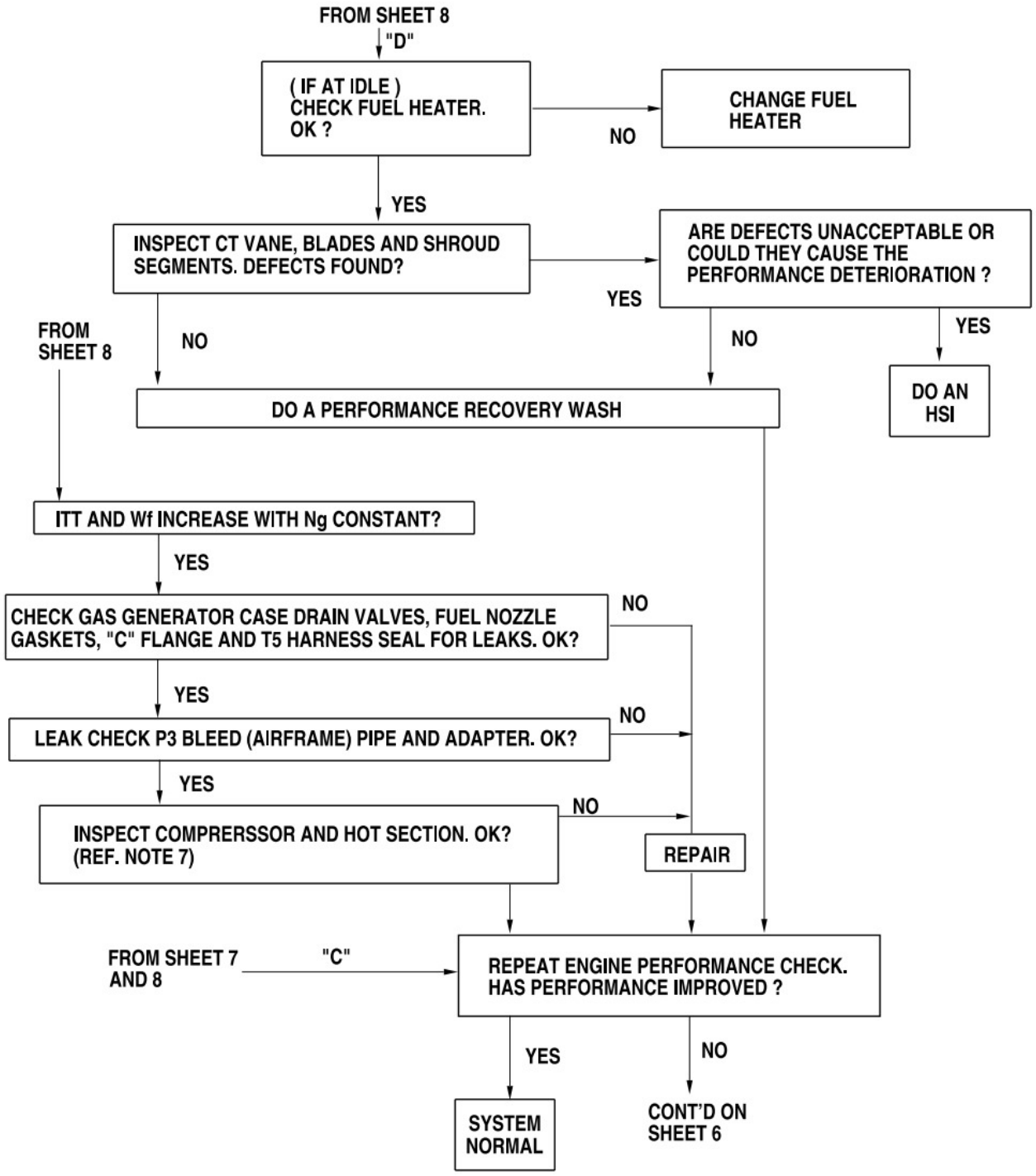
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NOTE: 1. Review ECTM data, pilot reports and maintenance log book entries and troubleshooting for the last 6 months. Hot section components and performance gradually deteriorate as running time accumulates. The deterioration rate varies according to operating conditions (environment and type of operation) and for different engine modification standards. If engine performance apparently improves T5 decrease without maintenance action, check indicating systems (usually T5 system is at fault).

NOTE: 2. A rapid shift in engine parameters is usually the result of an indicating system defect.

NOTE: 3. To extend hot section life, the following preventive maintenance, based on the increase in T5 from the values established at engine installation (engine performance/ground power check or ECTM), is recommended: For a 10°C (20°F) increase in T5, do a performance recovery wash. In addition, indicating system and engine performance/ground power checks are recommended to ensure reliable engine performance data. Also, test spray pattern or refurbish fuel nozzles.
For a 15°C (27°F) increase in T5 or a 1 to 1.5% decrease in Ng, do a borescope inspection of the combustion chamber, small exit duct, CT stator vanes and CT blades. In addition, do the maintenance recommended above.

NOTE: 4. A significant difference between ECTM and engine performance/ground power check parameters indicate that the defect is within the aircraft air bleed system on the affected engine (air bleed is off during an engine performance/ground power check). An alternative method is to do an engine performance/ground power check with air bleed ON then OFF. Compare the shift parameters for the engines on the same aircraft. A significant difference indicates bleed sharing problems. For single engine installations, contact P&WC for typical parameter shift recommendations.

NOTE: 5. When an engine is temperature limited on climb or cruise and the engine performance/ground power check is satisfactory, the defect is within the aircraft bleed system of the affected engine.

NOTE: 6. An increase in temperature without other parameter shifts may be the result of defective fuel nozzles or a deteriorated combustion liner altering combustion profile and changing the temperature distribution.

NOTE: 7. Compressor deterioration (which increases Ng) and hot section normal deterioration (which reduces Ng) may balance each other and the effect deterioration has on Ng will be very small or zero (i.e. Ng will remain constant).

NOTE: 8. When the engine is running on the ground, inlet plenum air pressure is lower than ambient air pressure. Therefore, if the inlet or firewall seals are damaged, hot air from around the engine may be ingested causing a temperature increase. Similarly, if the wind direction is from the rear, exhaust gas may be ingested and temperature increased.

NOTE: 9. The relationship between T5 and the temperature in front of CT stator may change due to hot section component deterioration altering combustion profile and/or gas path temperature distribution. This may affect the T5 trim and the indicated T5 and the relationship between T5 and the temperature in front of the CT vane used for engine certification. A T5 trim verification on the wing or in the test cell will confirm a shift in the indicated temperature.

NOTE: 10. To remove doubts, a T5 system check (including individual thermocouple checks) is recommended before doing a complete HSI.

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Table 101 ECTM Shift Fault Isolation Chart

ENGINE PARAMETERS	PROBABLE CAUSE	ACTION REQ'D	REMARKS
All parameters UP	AC/ engine indicating systems	Inspect	Repair as req'd
	Air inlet door blocked	Inspect	Repair as req'd
	Bypass door or ice vane mis-rigged.	Inspect	Repair as req'd
	Compressor FOD, rub, erosion	Inspect and/or repair	Remove engine if limits are exceeded
	Compressor contaminated/dirty	Wash compressor	
	Bleed valve open	Inspect	Repair as req'd
	P2.5 air leaks from engine/airframe system	Inspect	Repair as req'd
	PT stator vanes burned/flow area increased. PT blade tip oxidation/rub.	Inspect	Repair as req'd
	Hot start	Inspect hot section and check log book	Do applicable overtemperature inspection
All parameters DOWN	AC/engine indicating system	Inspect	Repair as req'd
Fuel nozzle deterioration	Inspect	Clean as req'd	
Ng UP or DOWN, T5, WF Constant.	AC/engine indicating system	Inspect	Repair as req'd.

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ENGINE PARAMETERS	PROBABLE CAUSE	ACTION REQ'D	REMARKS
Wf UP or DOWN, T5, Ng, constant.	AC/engine indicating system	Inspect	Repair as req'd.
Wf, T5 UP, Ng down	CT stator vanes burned/flow area increased. CT blade tip oxidation/rub.	Inspect CT stator, and CT blades.	If limits exceeded, do an HSI.
	Normal hot section deterioration	Do an HSI if T5 limit exceeded.	
Wf, T5 UP, Ng constant	P3 leaks from AC/engine system	Inspect	Repair as req'd
	Gas generator case cracked at fuel nozzle or P3 bleed pads. Diffuser exit ducts cracked or loose.	Inspect	Replace engine if defect confirmed.
	Leaking gas generator drain valves, C flange, fuel nozzle gasket, or T5 harness seal.	Inspect	Replace engine if defect confirmed.
	Concurrent hot section and compressor deterioration	Inspect CT stator and CT blades, and compressor.	Do an HSI or send engine for overhaul.

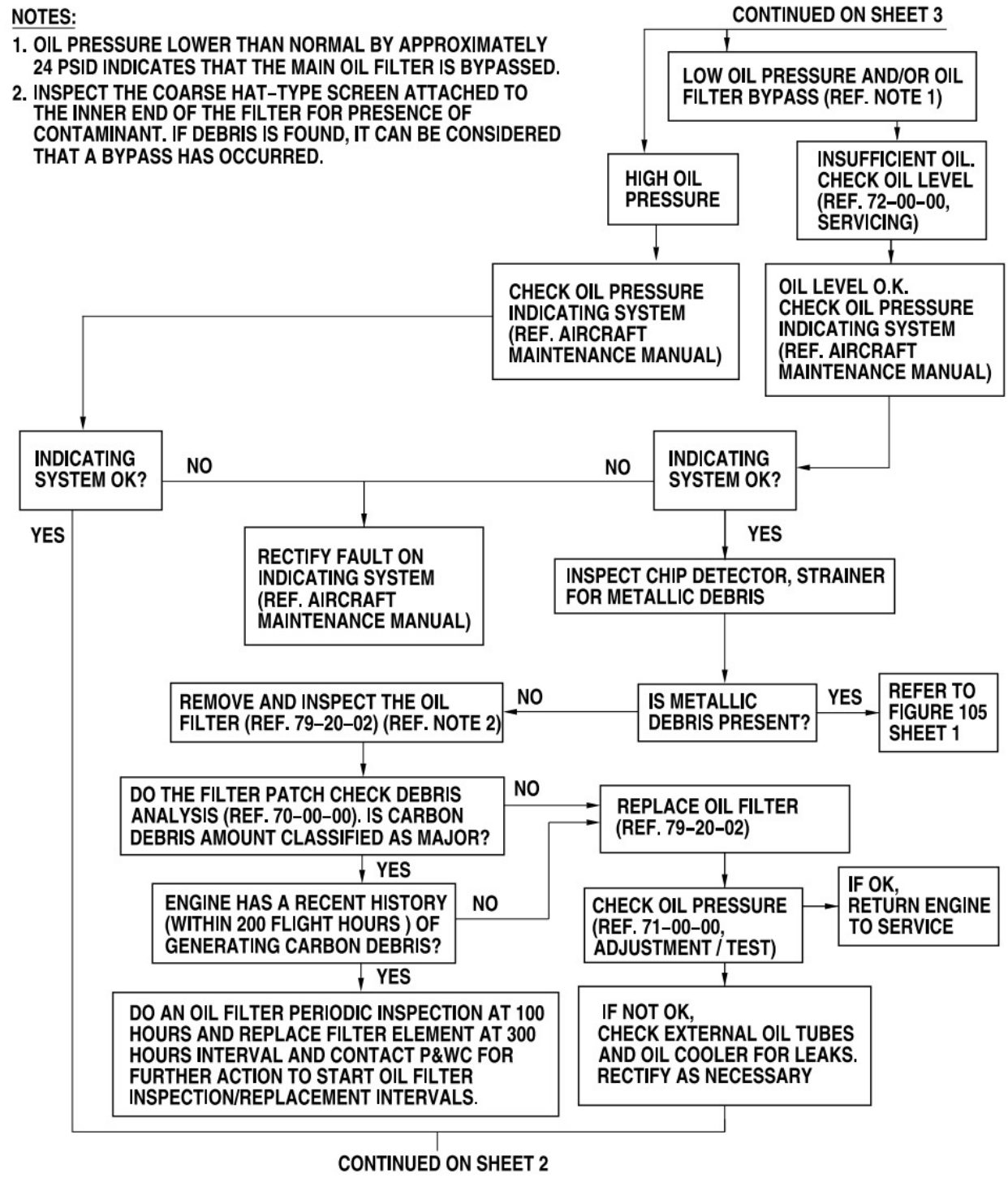
NOTE: 1. An increase in T5 without other parameter shifts may be the result of defective fuel nozzles or a deteriorating combustion chamber liner altering combustion profile and gas path temperature distribution.

NOTE: 2. The relationship between T5 and the temperature in front of the CT stator may change due to hot section component deterioration altering combustion profile and/or gas path temperature distribution. This may affect the T5 trim and the indicated T5, and the relationship between T5 and the temperature in front of the CT stator used for engine certification. A T5 trim verification on-wing or in a test cell will confirm a shift in the indicated temperature.

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Figure 104 Engine Lubrication Fault Isolation Chart

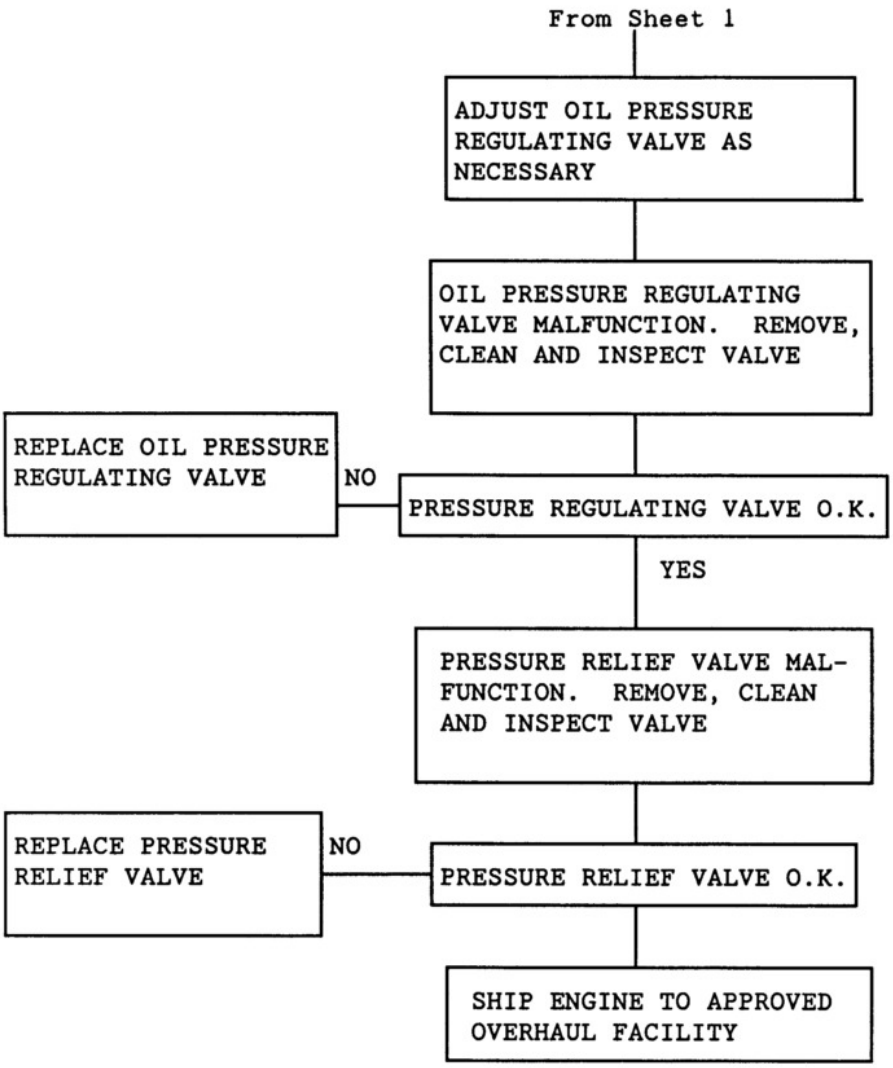
(SHEET 1 OF 9)



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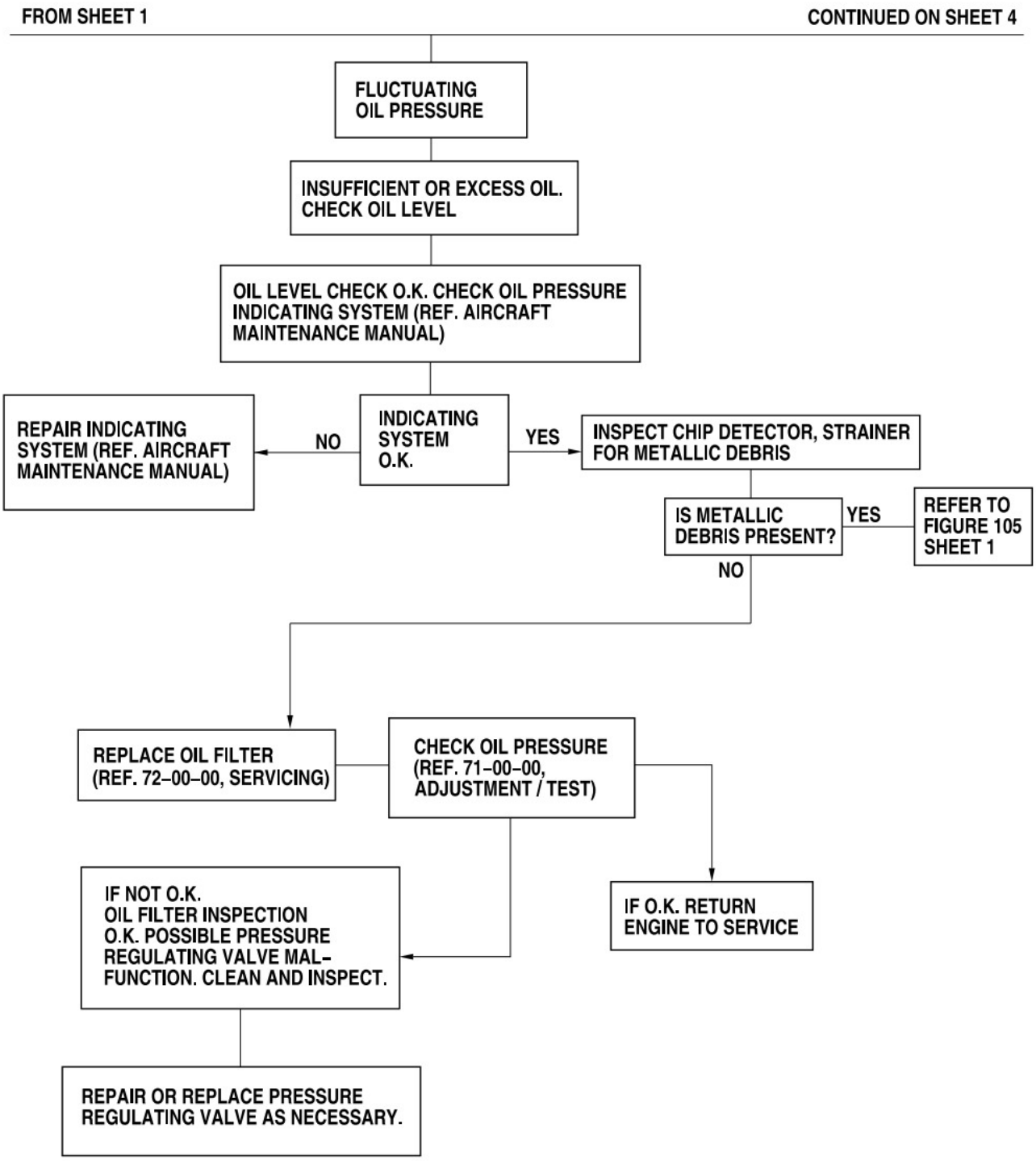
(SHEET 2 OF 9)



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(SHEET 3 OF 9)



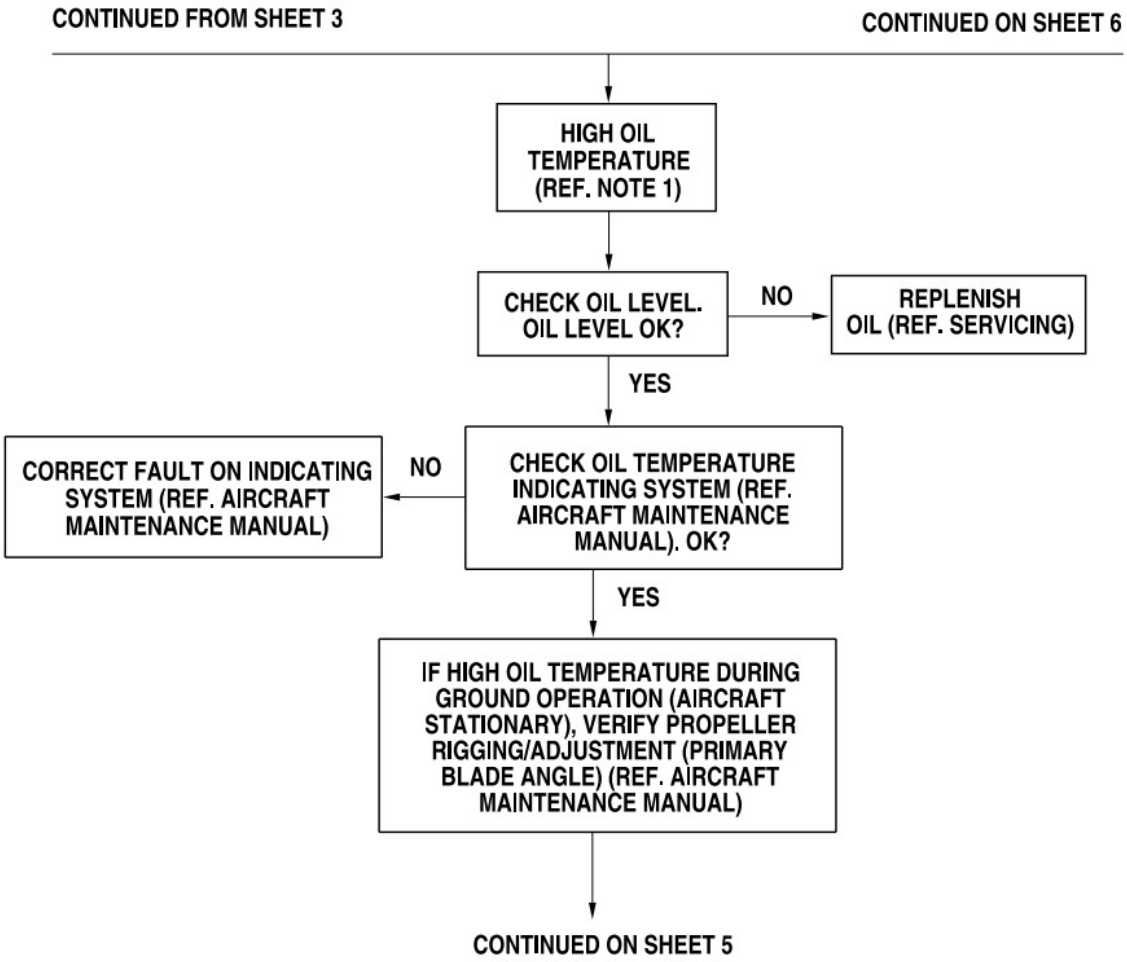
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(SHEET 4 OF 9)

NOTES:

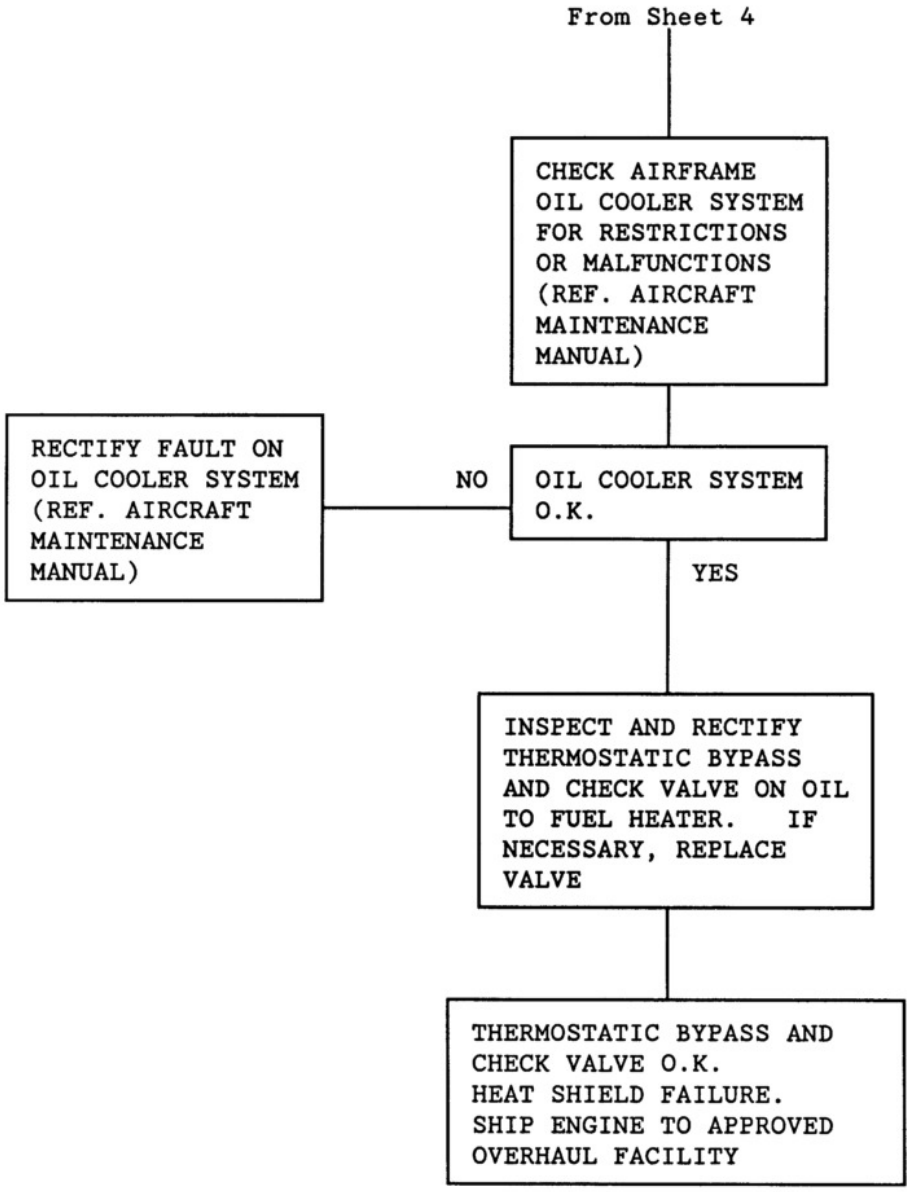
1. **EXCESSIVE IDLING IN FEATHER COULD CAUSE OVERHEATING INDICATION. RESTRICT IDLING TIME IN FEATHER.**
2. **IF THE ENGINE HAS BEEN OPERATED FOR AN EXTENDED PERIOD (REF. 71–00–00, POWER PLANT – ADJUSTMENT/TEST FOR TIME LIMIT) ABOVE OIL TEMPERATURE LIMITS, THEN DO AN UNSCHEDULED INSPECTION FOR SUSTAINED RUNNING AT OIL TEMPERATURE OUTSIDE LIMITS (REF. INSPECTION).**



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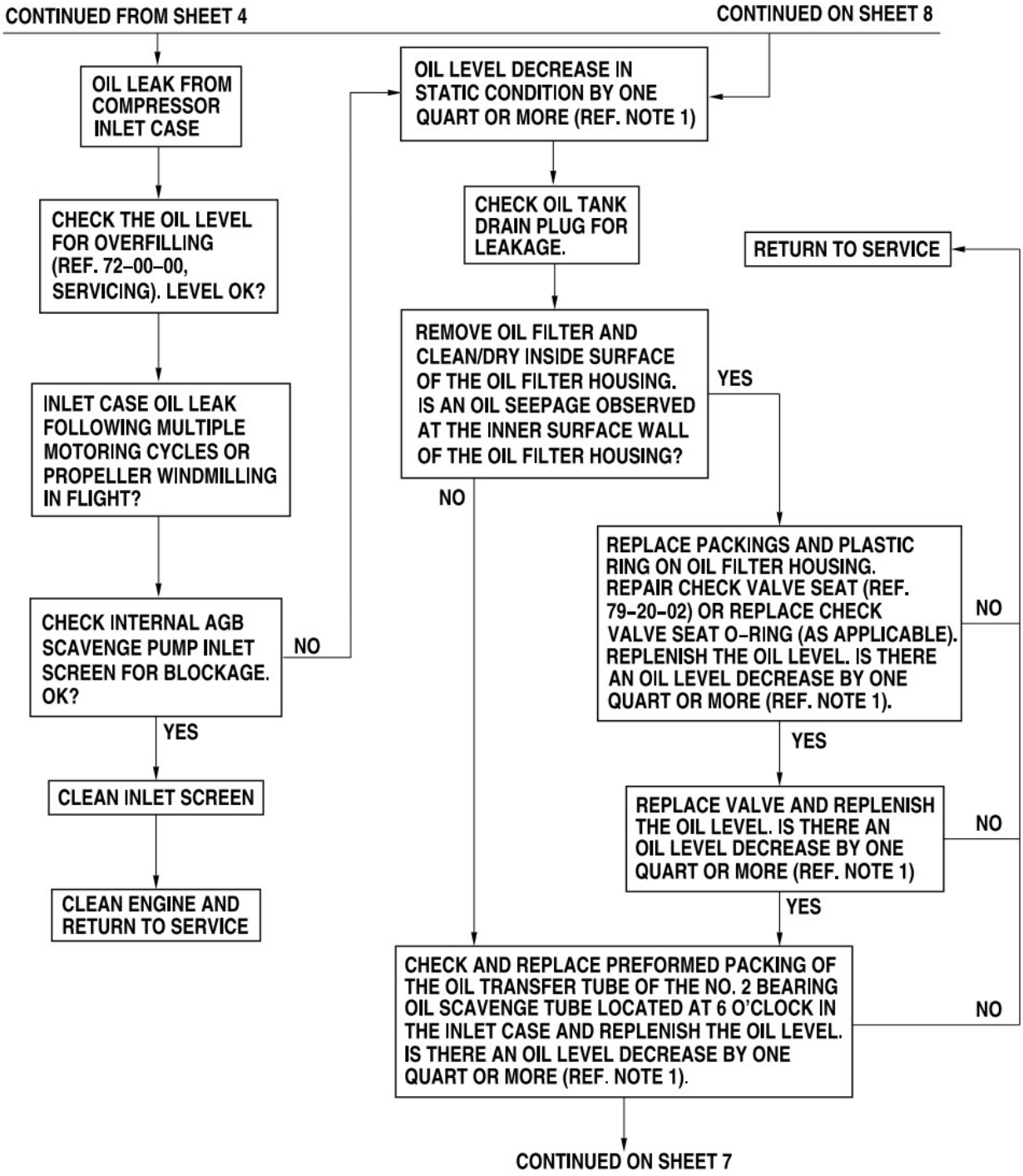
(SHEET 5 OF 9)



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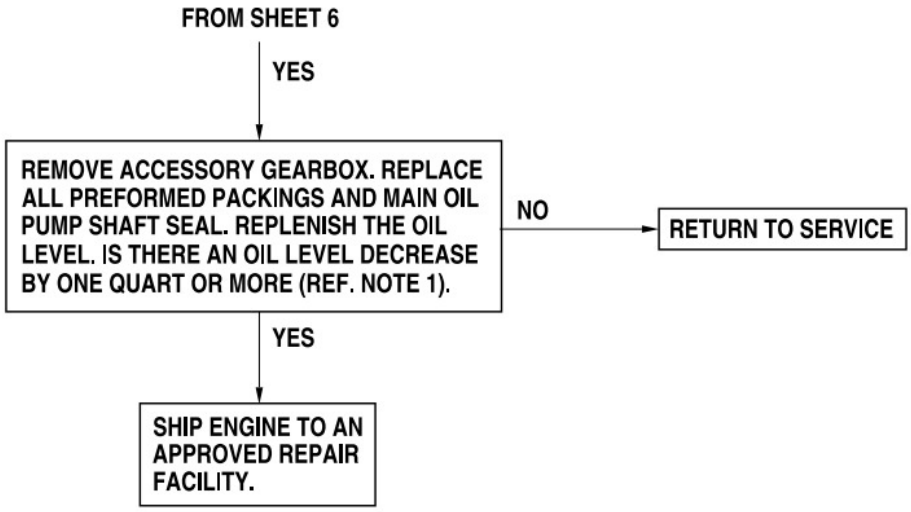
(SHEET 6 OF 9)



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(SHEET 7 OF 9)



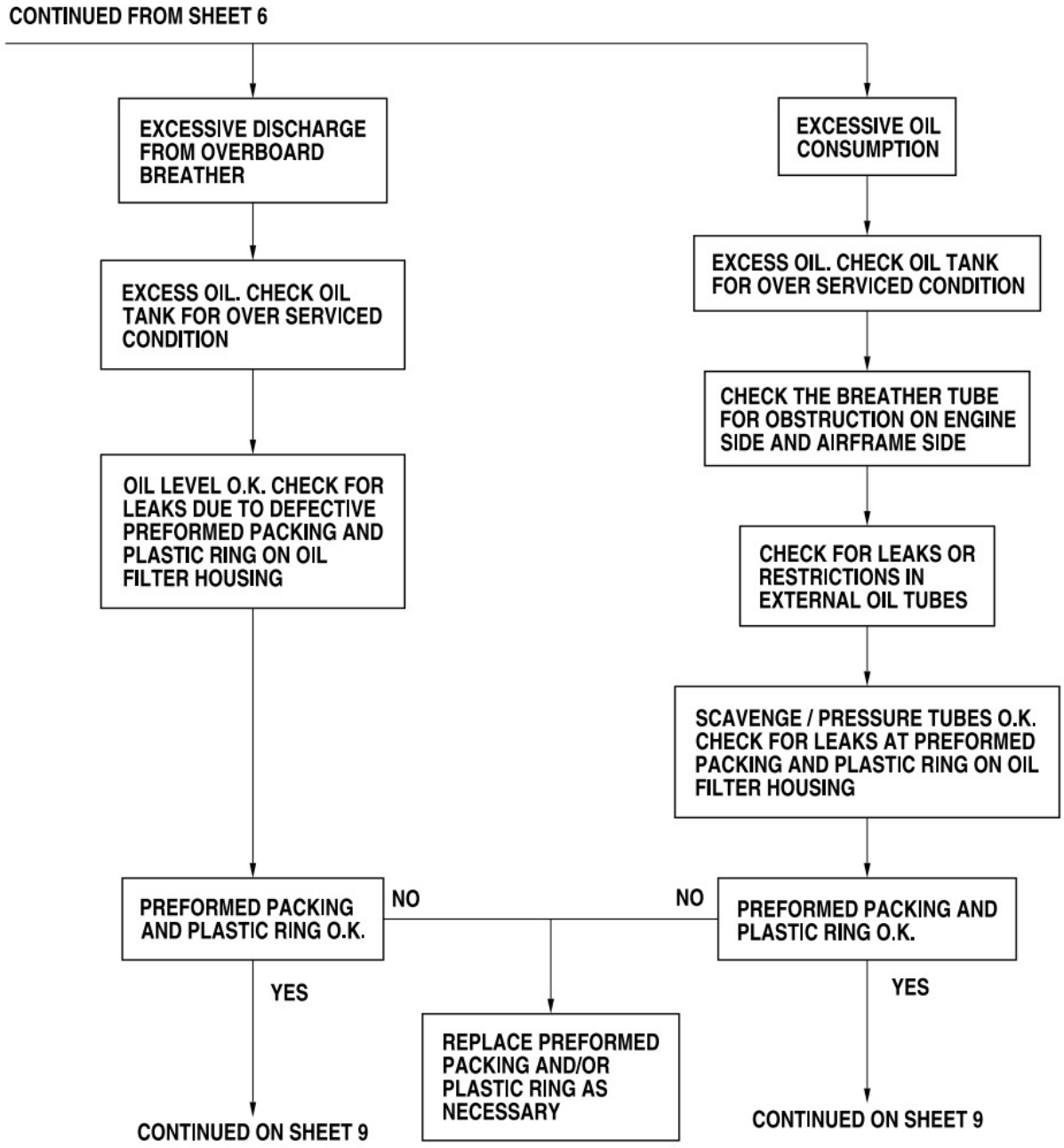
NOTE 1: A LEAK OF ONE QUART OR MORE AFTER 8 HOURS PERIOD (OVERNIGHT) MAY LEAD TO OIL OVER SERVICING AND ADDITIONAL OIL LEAKS ON VARIOUS LABYRINTH SEALS. FOR TROUBLESHOOTING PURPOSES AND FOR ACCURATE MEASUREMENT OF THE OIL STATIC LEAK, WE RECOMMEND TO PERFORM THE FOLLOWING STEPS:

(a) DRAIN THE ACCESSORY GEARBOX AND RGB CAVITIES. RE-INSTALL DRAIN PLUG.
(b) REPLENISH / SERVICE OIL LEVEL.
(c) VERIFY OR ESTABLISH BASELINE OIL LEVEL AFTER 15 MINUTES AND WAIT 8 HOURS TO VERIFY DECREASE OF OIL LEVEL.
(d) VERIFY THE OIL LEVEL TO IDENTIFY IF THE LEAKAGE IS STILL MORE THAN ONE QUART OVER 8 HOURS PERIOD.

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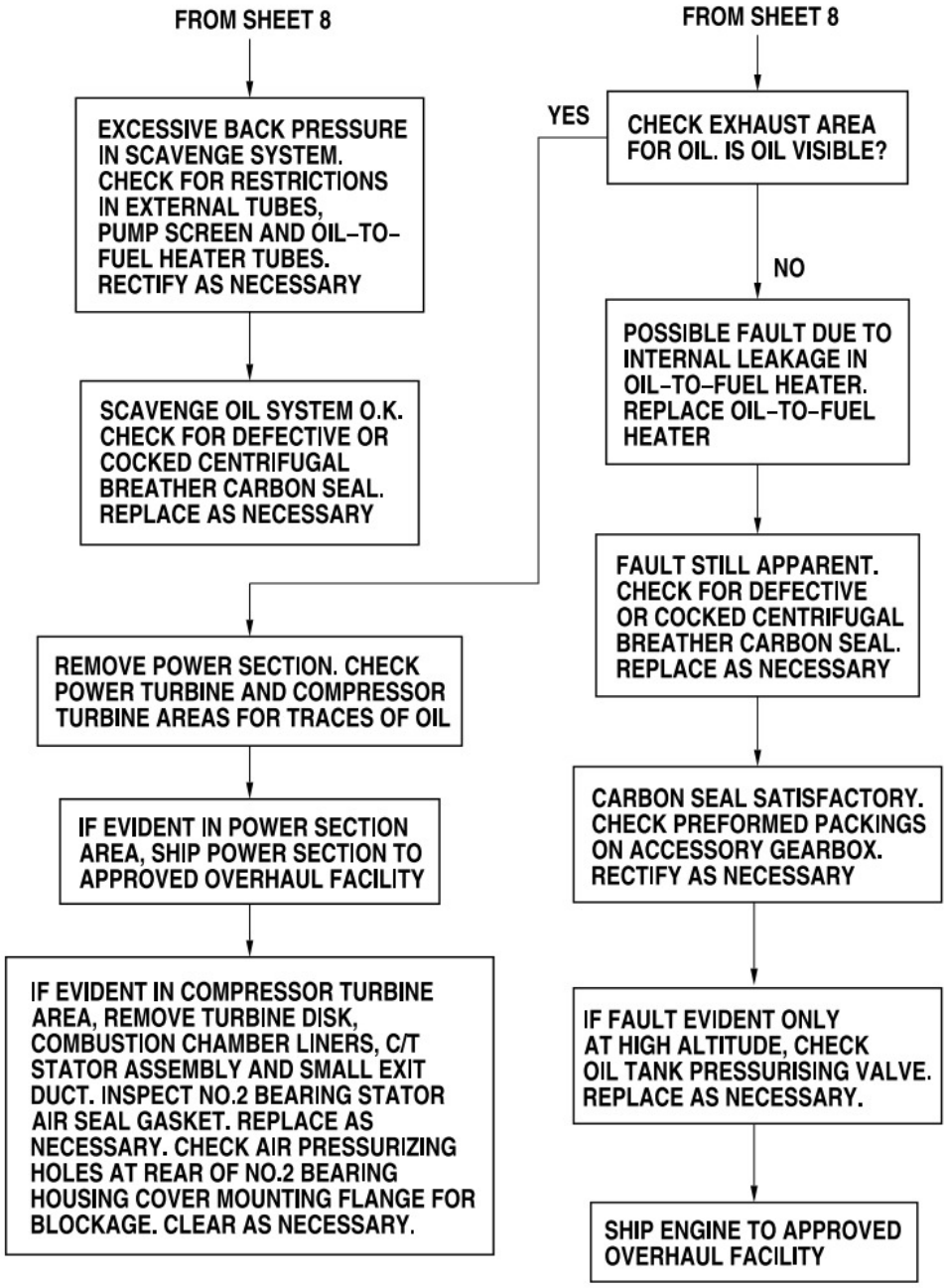
(SHEET 8 OF 9)



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(SHEET 9 OF 9)

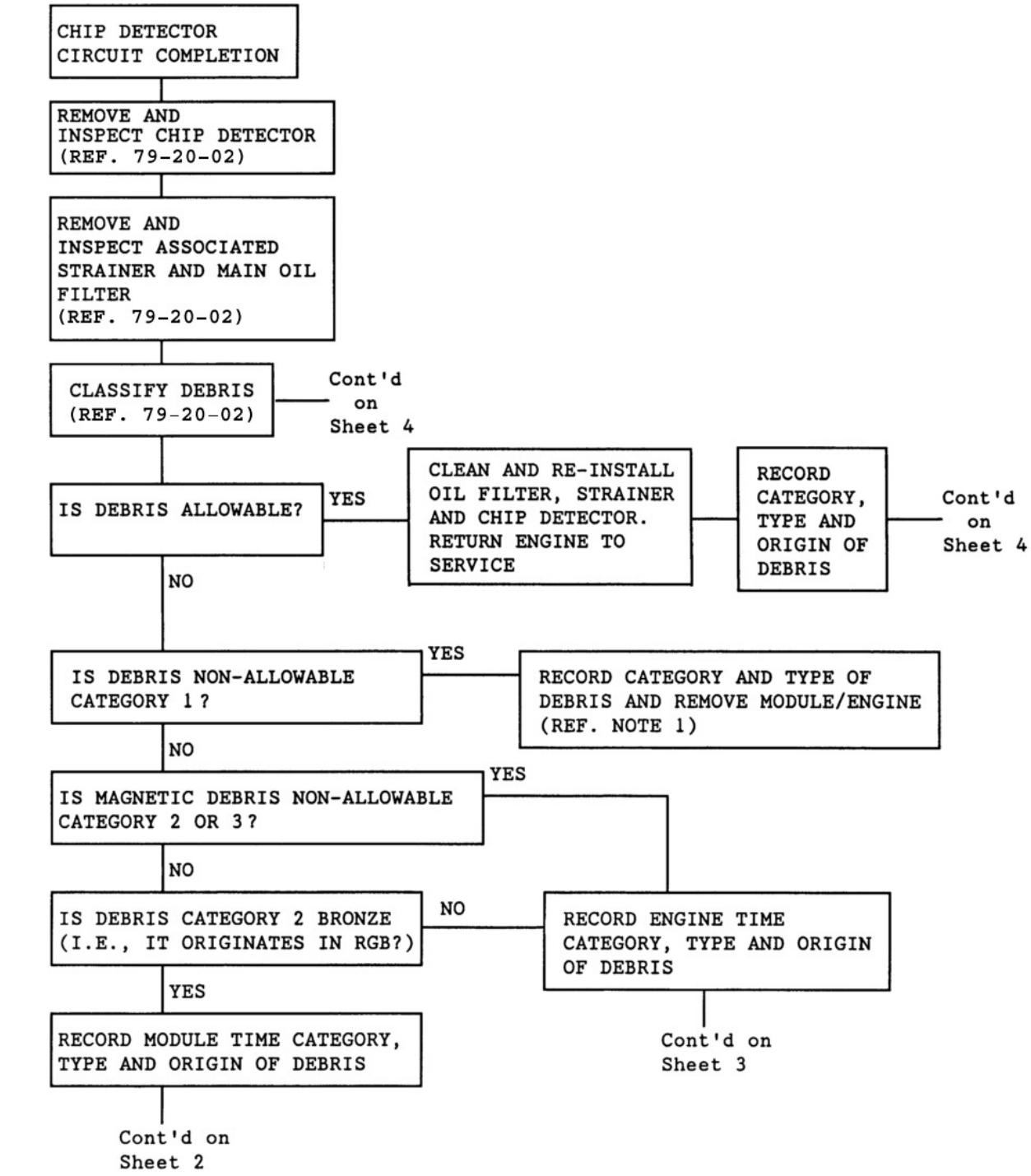


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Figure 105 Engine Lubricating Oil Contamination Fault Isolation Chart

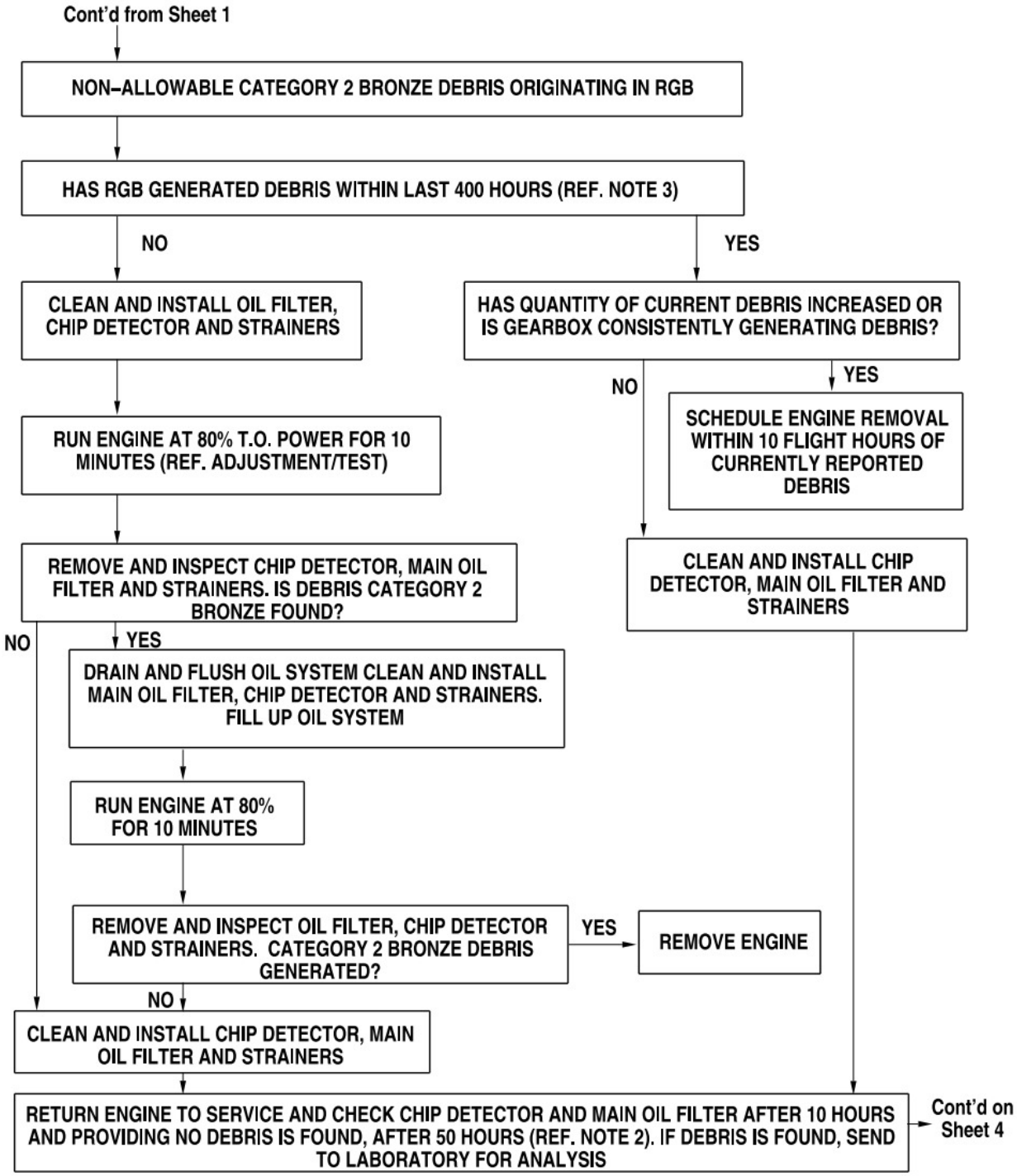
(SHEET 1 OF 4)



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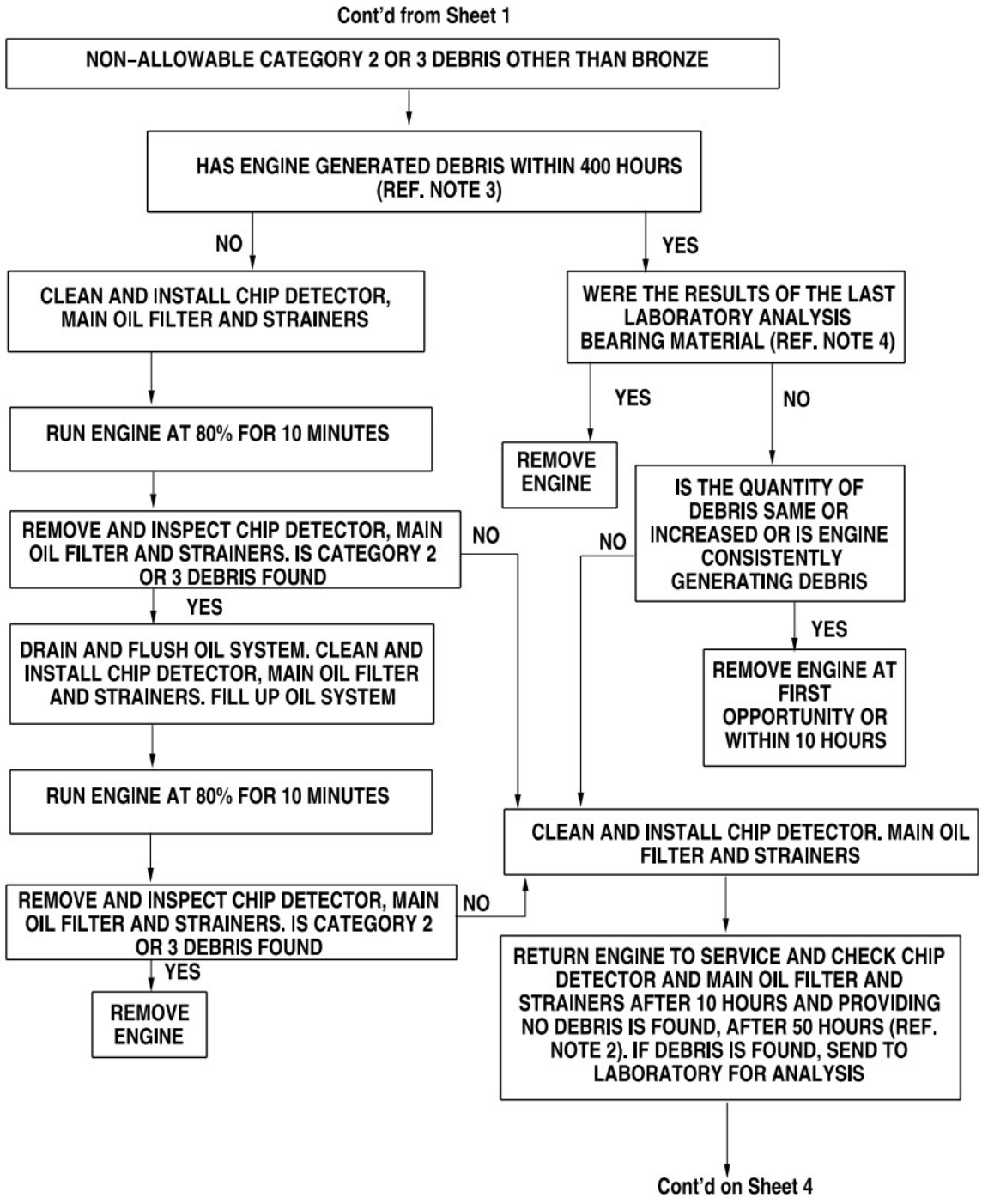
(SHEET 2 OF 4)



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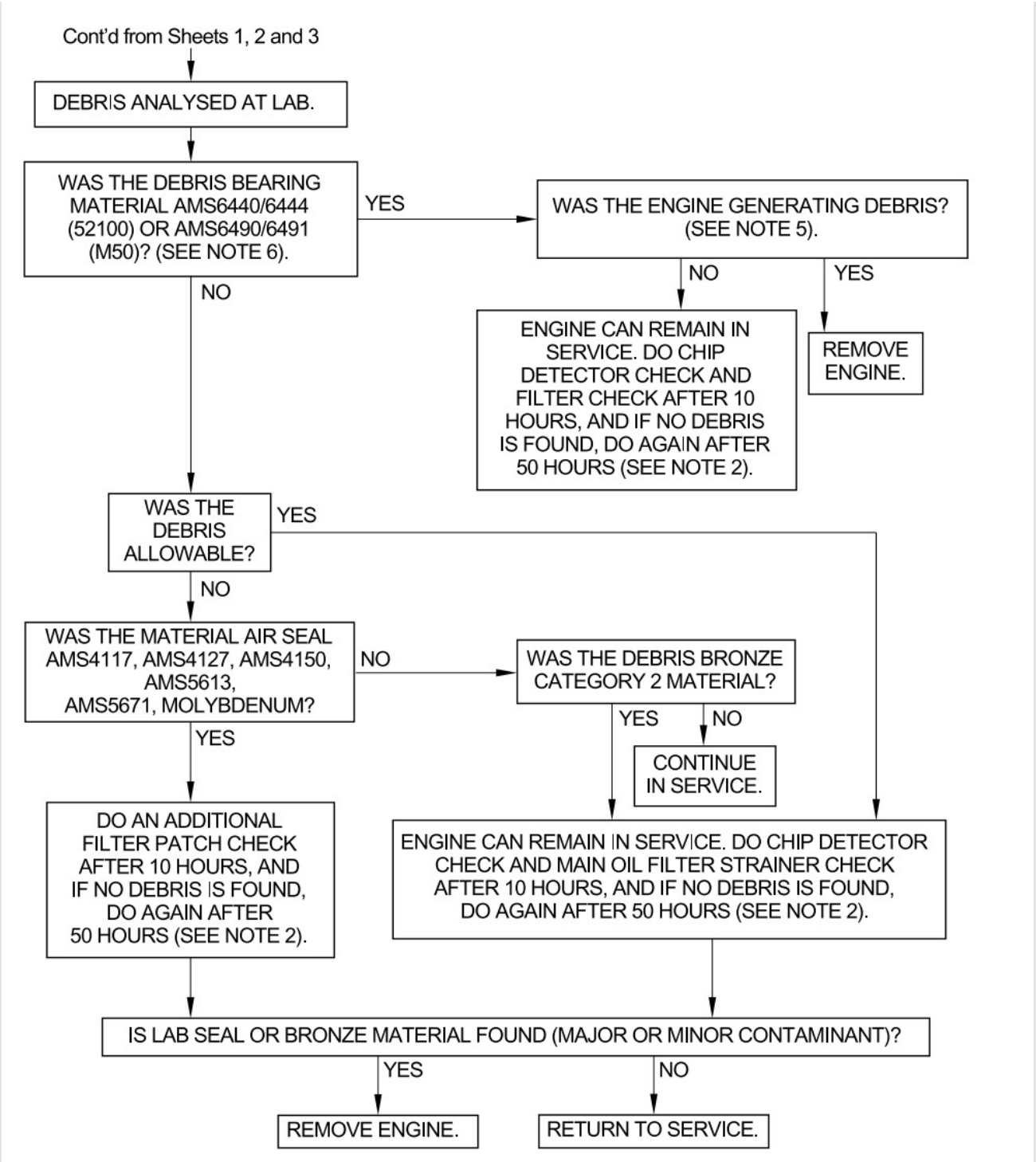
(SHEET 3 OF 4)



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(SHEET 4 OF 4)



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(2) The following NOTES are referred to in Figure 105:

NOTE: 1. Depending where the debris was found (i.e., strainer, main oil filter or chip detector) and the shape (i.e., tooth part, locking washer part etc.) a decision could be made concerning the removal of the affected module. Alternatively, remove engine.

NOTE: 2. If no debris is found at the 50-hour check, return the oil filter and chip detector inspections to operator's standard maintenance program intervals.

NOTE: 3. Results of previous laboratory analysis and origin of debris should be determined within 50 hours of original detection of debris.

NOTE: 4. Results of last sample must be known prior to continuing.

NOTE: 5. It should have already been determined that the engine and not RGB mounted accessories are generating the debris. If propeller governor or O/S governor failure occurred before the original debris detection and the material could be identified as a propeller or O/S governor material then the engine may not be generating the debris and therefore could remain in service, if the components of the propeller governor or O/S governor are accounted for.

NOTE: 6. Bearing inner and outer rings as well as rolling elements are made of steel AMS 6440/6444 (52100) or AMS 6490/6491 (M50). Bearing cages are made of steel AMS 6414/6415 (4340) or bronze AMS 4616, silver plated per AMS 2412. When only bearing cage material is generated (major or minor filter contaminant), this may indicate cage wear due to lack of lubrication and/or bearing damage causing rolling element instability.

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